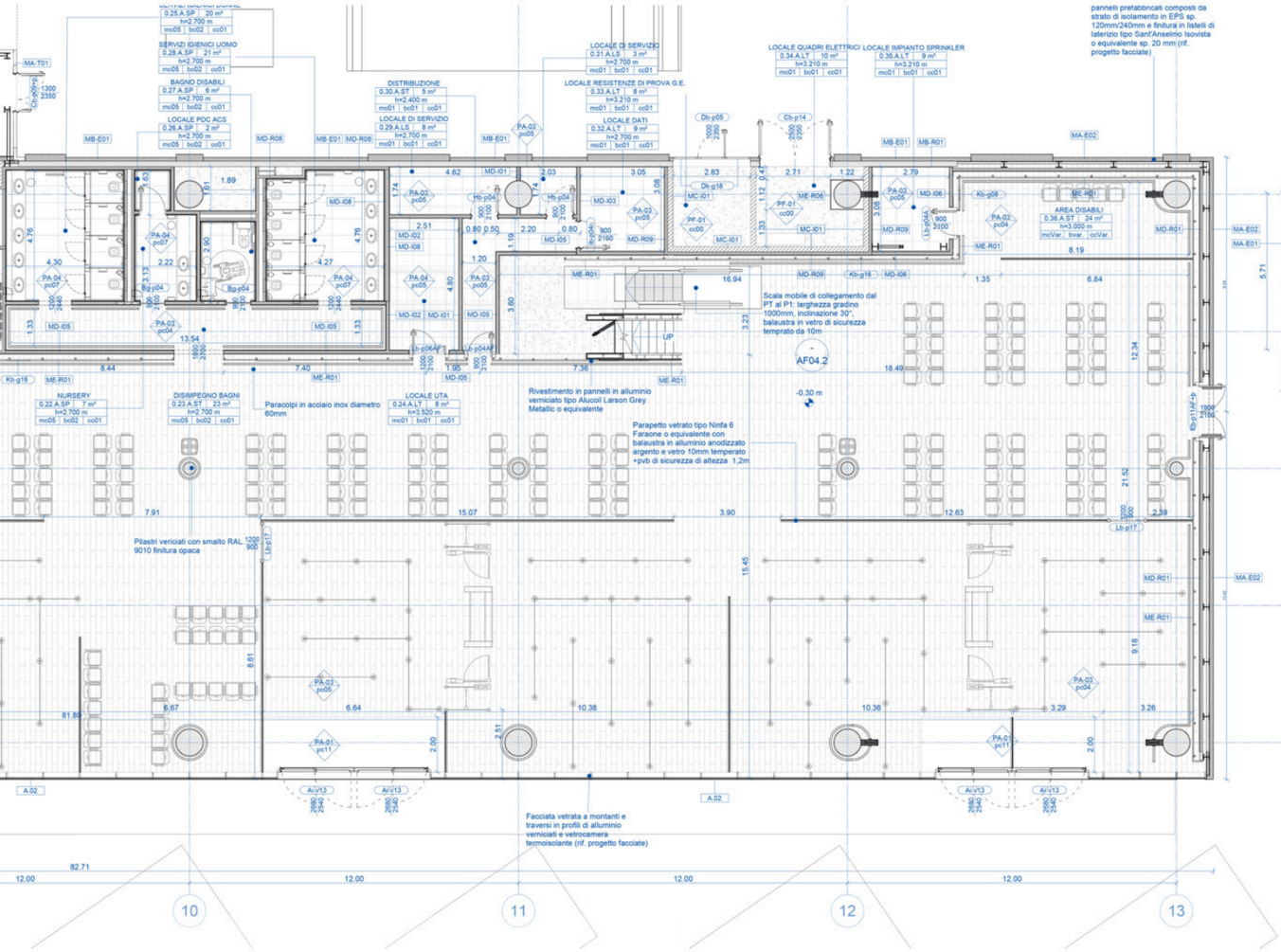
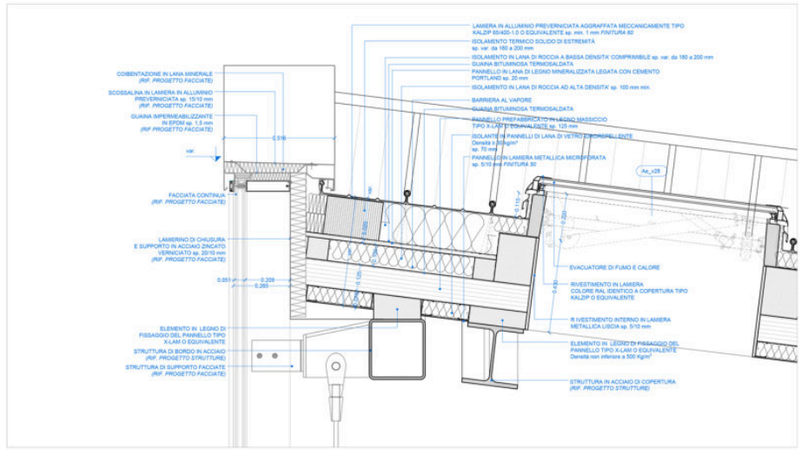


SELECTED WORKS 2022–2026

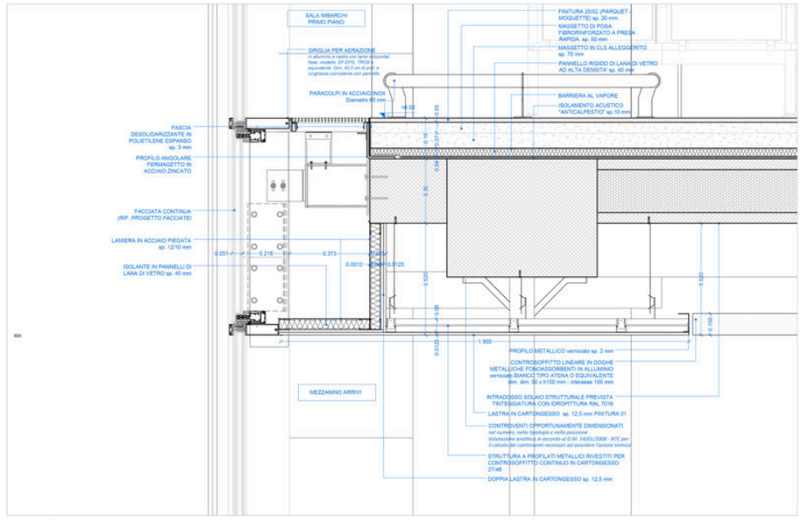
PORTFOLIO

MARCELLO SEBASTIANI ARCHITECT AND CIVIL ENGINEER

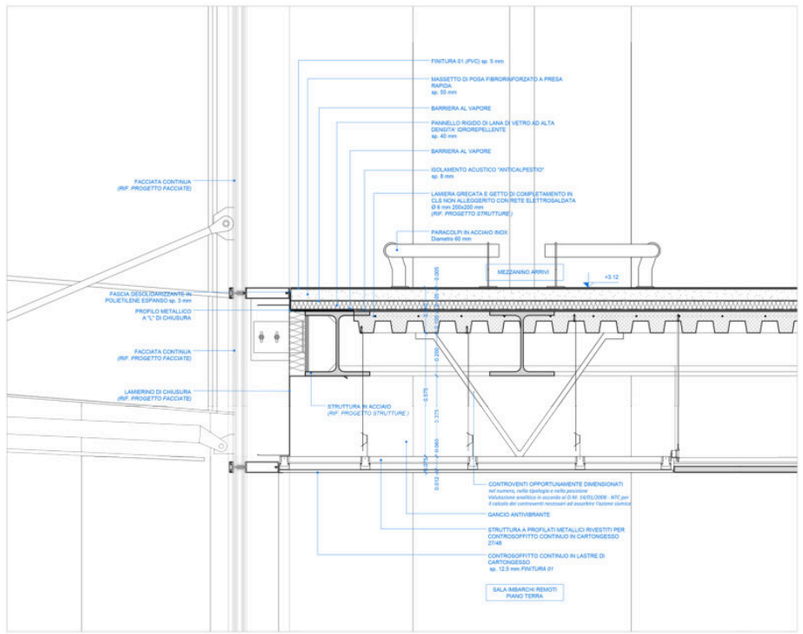
mar.sebastiani.thai@gmail.com · +66 82 846 1209 · Bangkok, Thailand



01 Dettaglio scossalina facciata continua airside nord-est
1:10



02 Dettaglio controsoffitto piano mezzanino facciata airside nord-est
1:10



03 Dettaglio controsoffitto sala imbarchi remoti -piano terra
1:10



MARCELLO SEBASTIANI

ARCHITECT AND CIVIL ENGINEER

BASED IN

Bangkok, Thailand

REGISTRATION

Registered with the Order of Engineers of Rome

EDUCATION

Sapienza Università di Roma; Single-cycle M.Sc. Building-Architecture Engineering (LM-4), 2017–2023

STUDIO

Co-Founder & Architectural Project Manager, Kolibri Studio; kolibri.global

CONTACT

mar.sebastiani.thai@gmail.com
+66 82 846 1209

ABOUT

I have worked on six projects spanning data-centre technical infrastructure, a university library design competition, defence and humanitarian facilities, and two major airport terminal expansions. Across all of them I have worked with the mindset of both engineer and designer, handling everything from structural systems and construction detailing through to BIM coordination.

My career has been a fortunate one: I earned trust early, along with the opportunity to work directly on construction sites. Working closely with every kind of supplier and subcontractor, with executing contractors, designers and site management teams, I learned early to design with a multidisciplinary awareness, attending to every aspect down to the detail of the finishes.

It is this solid engineering foundation that I now want to put to the service of a more conscious and sustainable design, grounded in carefully weighed choices. Increasingly, that also means adopting a new approach to construction itself: using advanced software and more accurate modelling techniques to catch potential issues at the design stage, reducing costs and timelines in favour of quality, sustainability and efficiency.

The other design objective is to integrate architecture into its surrounding context; respecting the site while balancing sustainability and comfort, without sacrificing one for the other. An aesthetic pursuit always in dialogue with the different contexts, cultures and characteristics of each site.

CONTENTS

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01 **Avalon-3 Data Center** — Civil works for a data-centre campus, client Irideos — p.04

02 **Nuova Biblioteca Lorenteggio** — International library design competition — Sapienza Università di Roma — p.06

03 **Integra-AES — Studio Collaboration** — Humanitarian, institutional and defence-sector commissions — p.10

04 **Verona Villafranca Airport** — Departures terminal expansion — construction site technical office — p.12

05 **Venice Marco Polo Airport** — New terminal — site technical office, BIM Manager & Coordinator — p.14

06 **The White Lotus Koh Samui** — Kolibri Studio flagship project — residential condominium, Koh Samui — p.18

Chapter 01

Avalon-3 Data Center

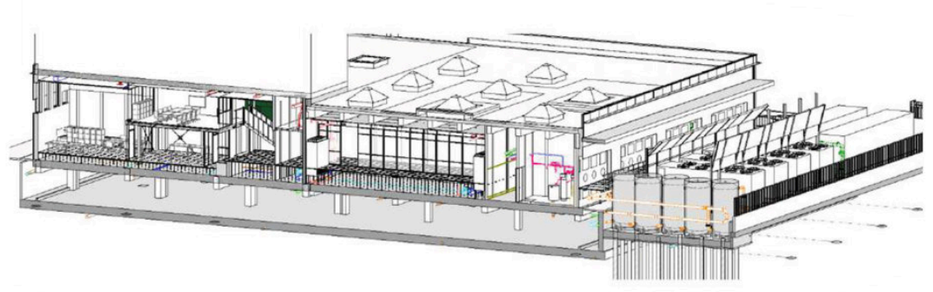
Civil engineering scope of the Avalon-3 data-centre construction project for client Irideos, through the BIM consultancy Bimon S.r.l.: first as BIM Specialist, supporting the site as the sole point of contact for my firm; then as BIM Coordinator, developing the construction project for the civil works. Working with the Project Management team, suppliers and contractors, I covered metalwork, reinforced-concrete works, floating floors and fire protection systems, designed parts of the facility including the wastewater treatment plant, and coordinated two BIM specialists.

LOCATION
Milan, Italy

CLIENT
Irideos; Data Center

CONSULTANCY
Bimon S.r.l., Rome

PERIOD
February – November 2022



BIM coordination model — civil, structural and MEP

TECHNICAL NOTES; ENGINEERING SCOPE

STRUCTURAL SYSTEM

Reinforced-concrete pad foundations for tanks, chiller units and the MV/LV substation. Office mezzanine with a steel structure on HEB180 columns and beams, stiffened with ø20 tie-rods on hinged flange connections. Frame structures supporting generator sets and similar equipment, and elevated walkways. Support structures for busbar ducts.

FLOOR SYSTEM

Modular floating (raised access) floor system.

WALL SYSTEM

REI-certified block walls with horizontal and vertical reinforced stiffening.

ACOUSTIC ABSORPTION SYSTEM

Redesign of the sound-absorbing walls and their support structure, coordinated with the chiller unit and tank plant complex.

FIRE SAFETY

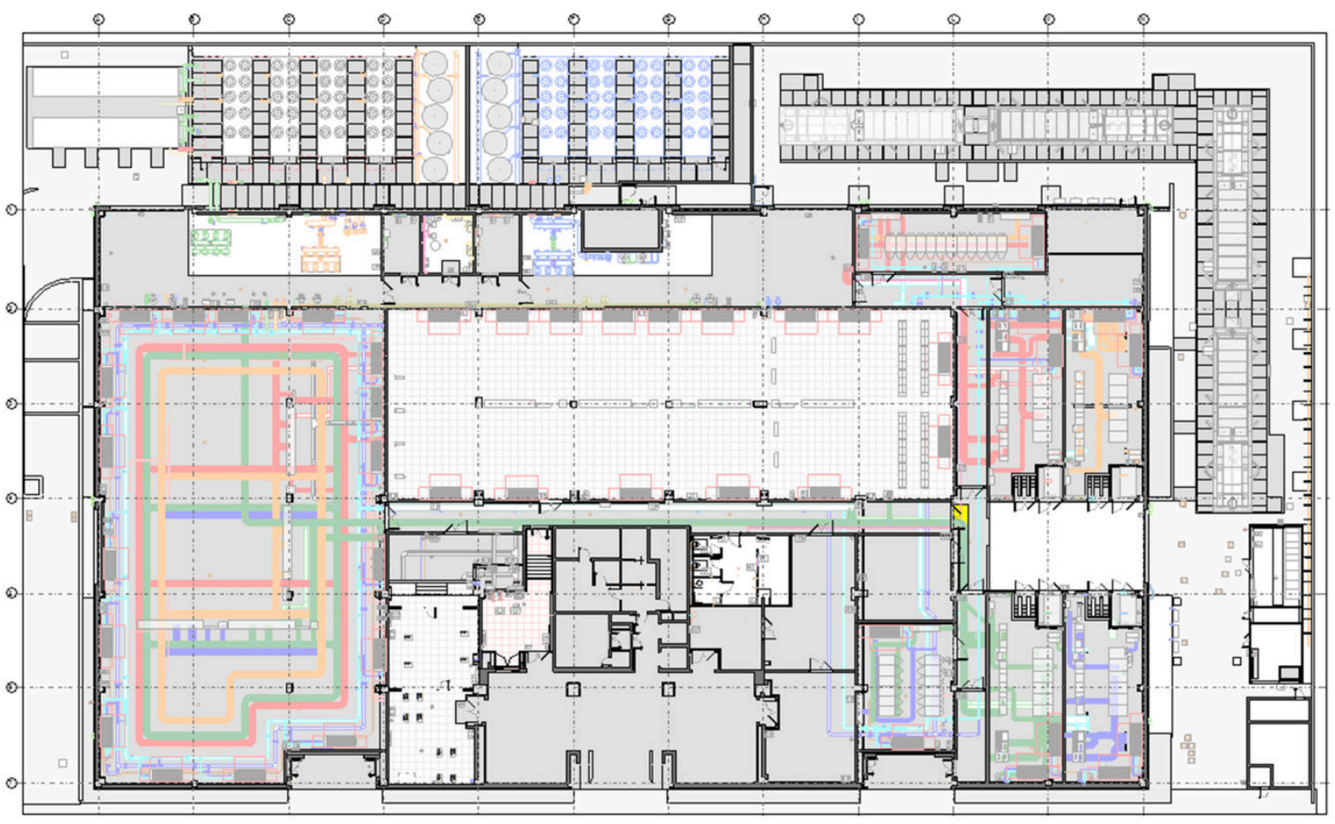
REI compartmentation coordinated across the entire civil works scope.

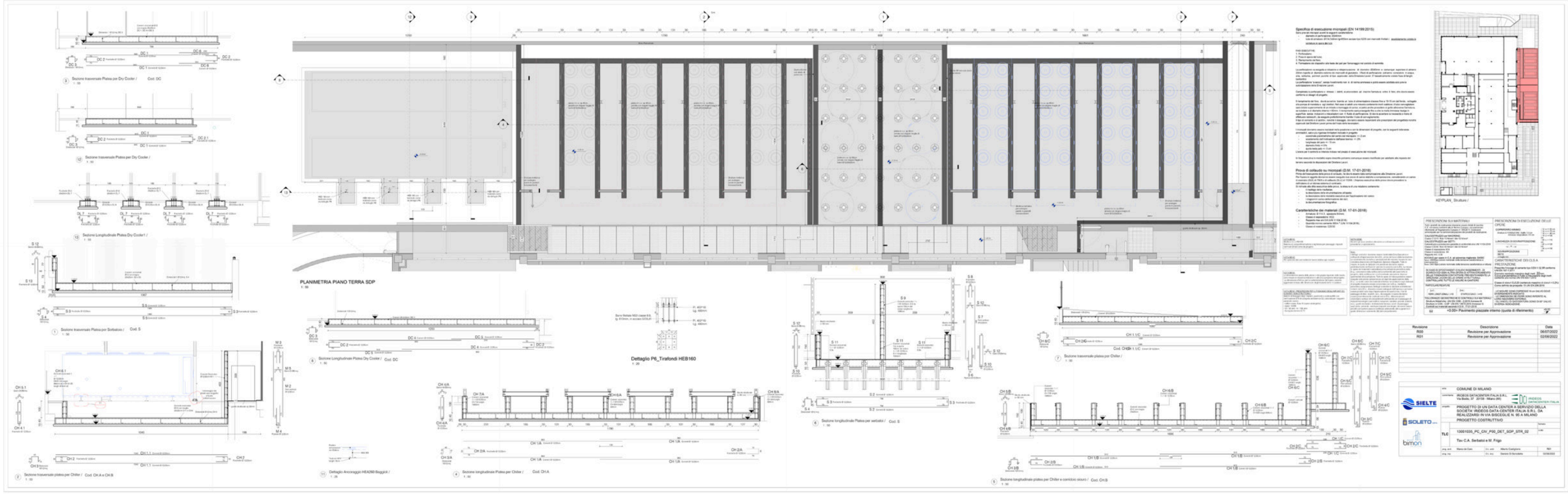
PLANT ENGINEERING

Designed parts of the wastewater treatment plant and the underground cabling network.

STANDARDS

UNI 11104:2016 (high-performance concrete) · UNI EN 933-1:2012 (fill aggregate grading).





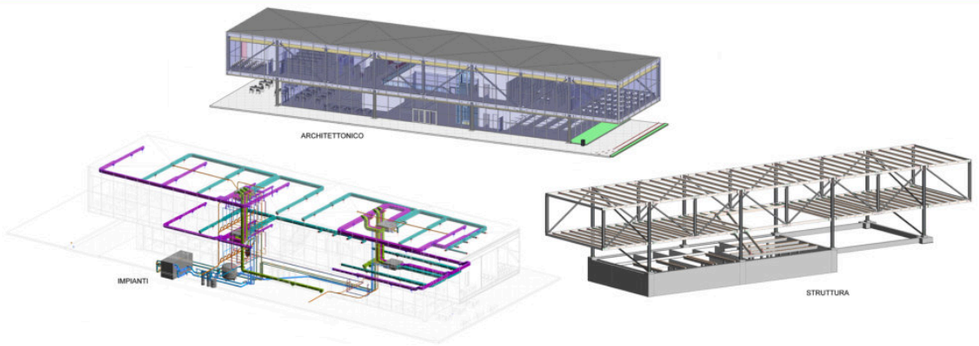
Tanks & chiller units — reinforced-concrete details

13001035_PC_CIV_P00_DET_SDP_STR_02



Office mezzanine — general plan, sections & details

13001035-PC-CIV-P00-DET-SDP-UFF-07



Exploded axonometric — architecture / structure / MEP

Chapter 02

Nuova Biblioteca Lorenteggio

Master's thesis developed for the international design competition “Nuova Biblioteca Lorenteggio”; a new library in Milan. The project was carried across three integrated disciplines: architecture, structure and building services (MEP).

INSTITUTION

Sapienza Università di Roma

PROGRAMME

M.Sc. Ingegneria Edile-Architettura; A.Y. 2022/23

SUPERVISORS

Rel. ing. Marina Pugnaletto · Co-rel. ing. S. Perno, ing. S. Grignaffini

CONTEXT

International Design Competition; Milan, Italy

TECHNICAL NOTES; ENGINEERING SCOPE

STRUCTURAL SYSTEM

Sizing, calculation and modelling of the primary steel frame structure, with braced bays, connection joints, stringer staircases and foundations with a below-grade reinforced-concrete level. Developed under the supervision of ing. Salvatore Perno (Tecnica delle Costruzioni).

MEP COORDINATION

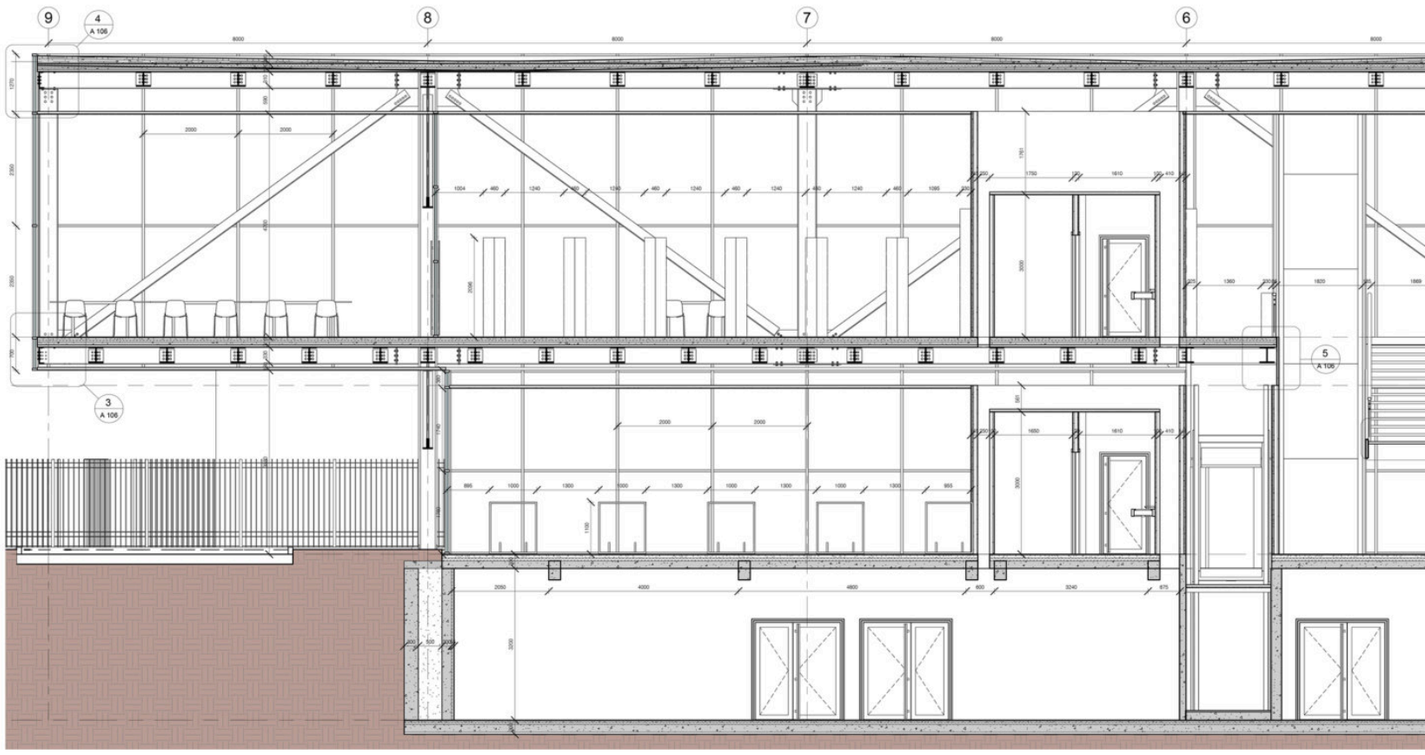
Thermal load calculation, HVAC system sizing, hydraulic and drainage networks, electrical containment and plant rooms coordinated across all levels; with ing. Stefano Grignaffini (Impianti Termo-Tecnici).

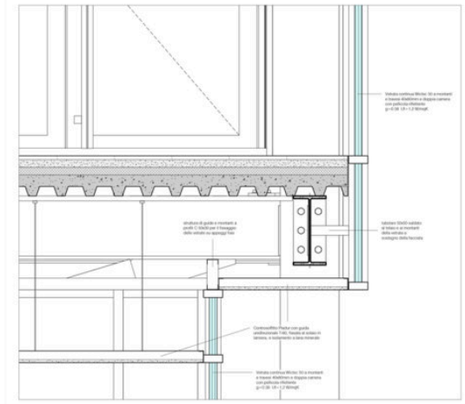
METHOD

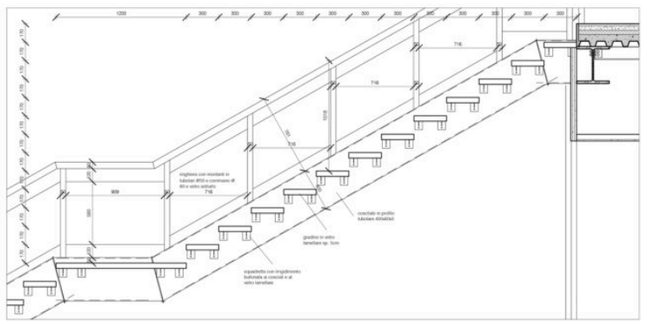
Architecture, structure and MEP modelled as three federated disciplines and presented together (see exploded axonometric, plate A100).

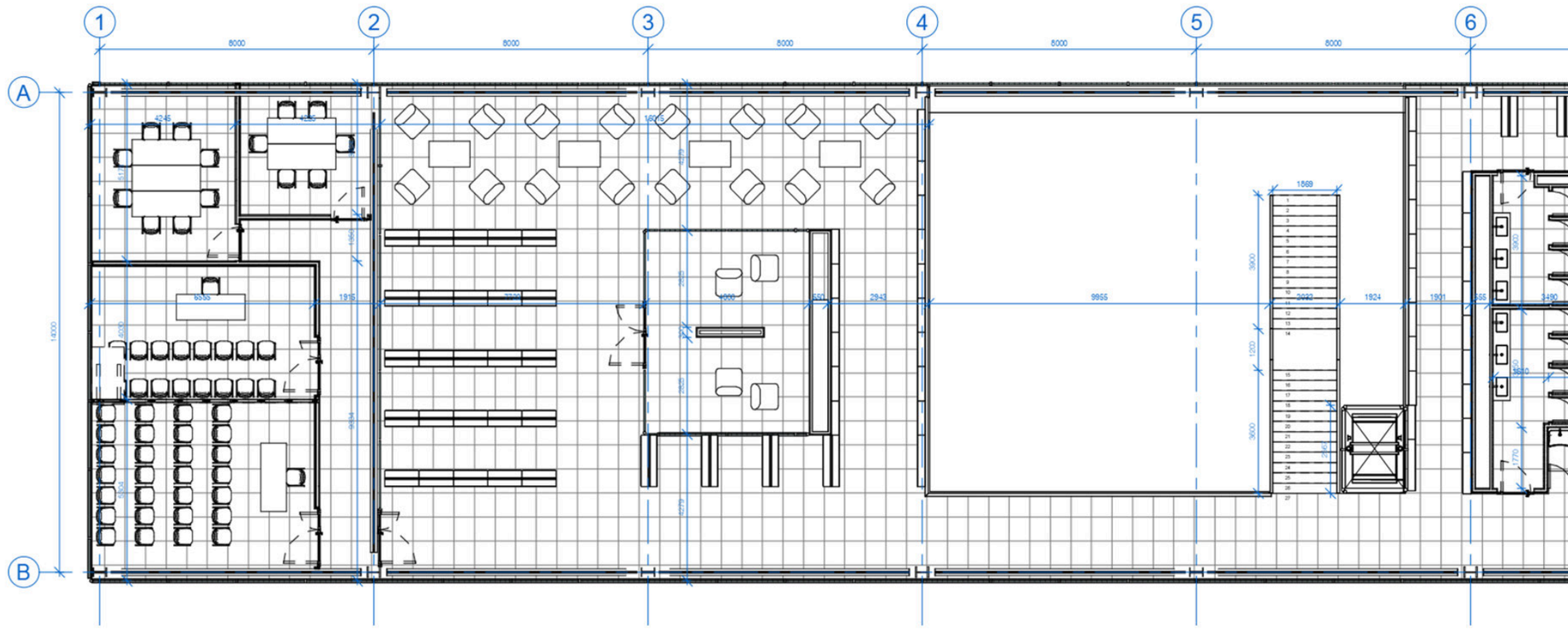
STANDARDS

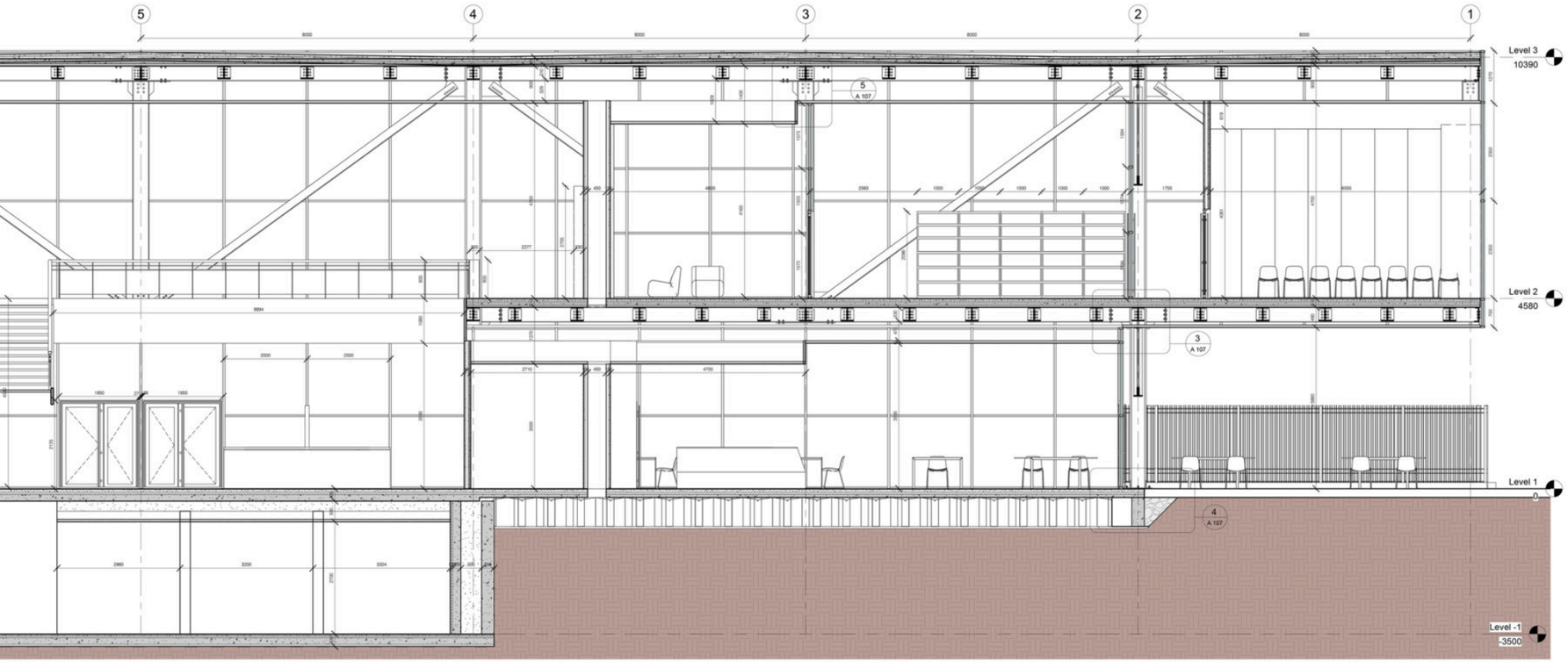
Design of publicly accessible places and accessibility (UNI EN 17161:2019 · UNI EN 17210).

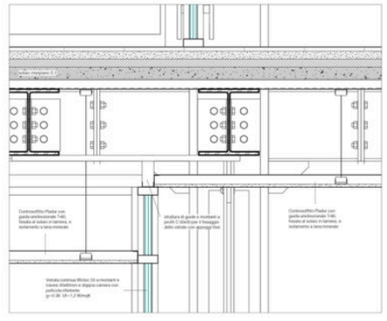


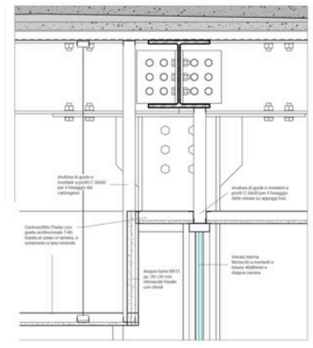


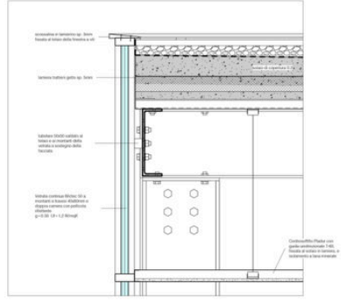


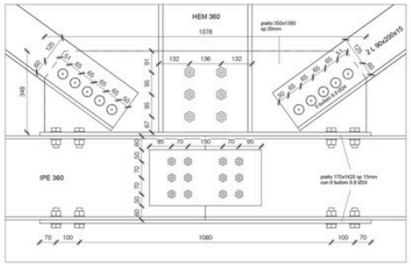

1
2
3
4
5
6
A
B
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14000
1868
3600
1220
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1901
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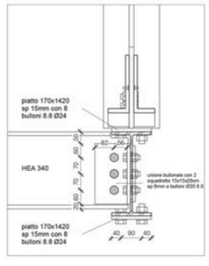

5
4
3
2
1
8000
Level 3
10390
Level 2
4580
Level 1
0
Level -1
-3500
5
A 107
3
A 107
4
A 107
9994
2000
2000
1950
1950
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3304
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2583
1000
1000
1000
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2710
4700

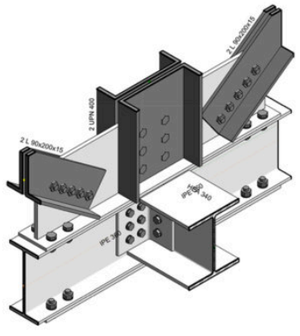


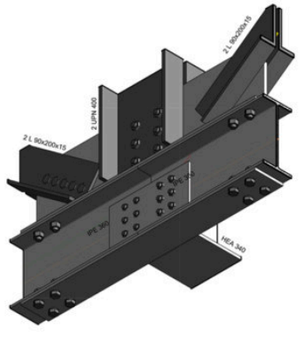


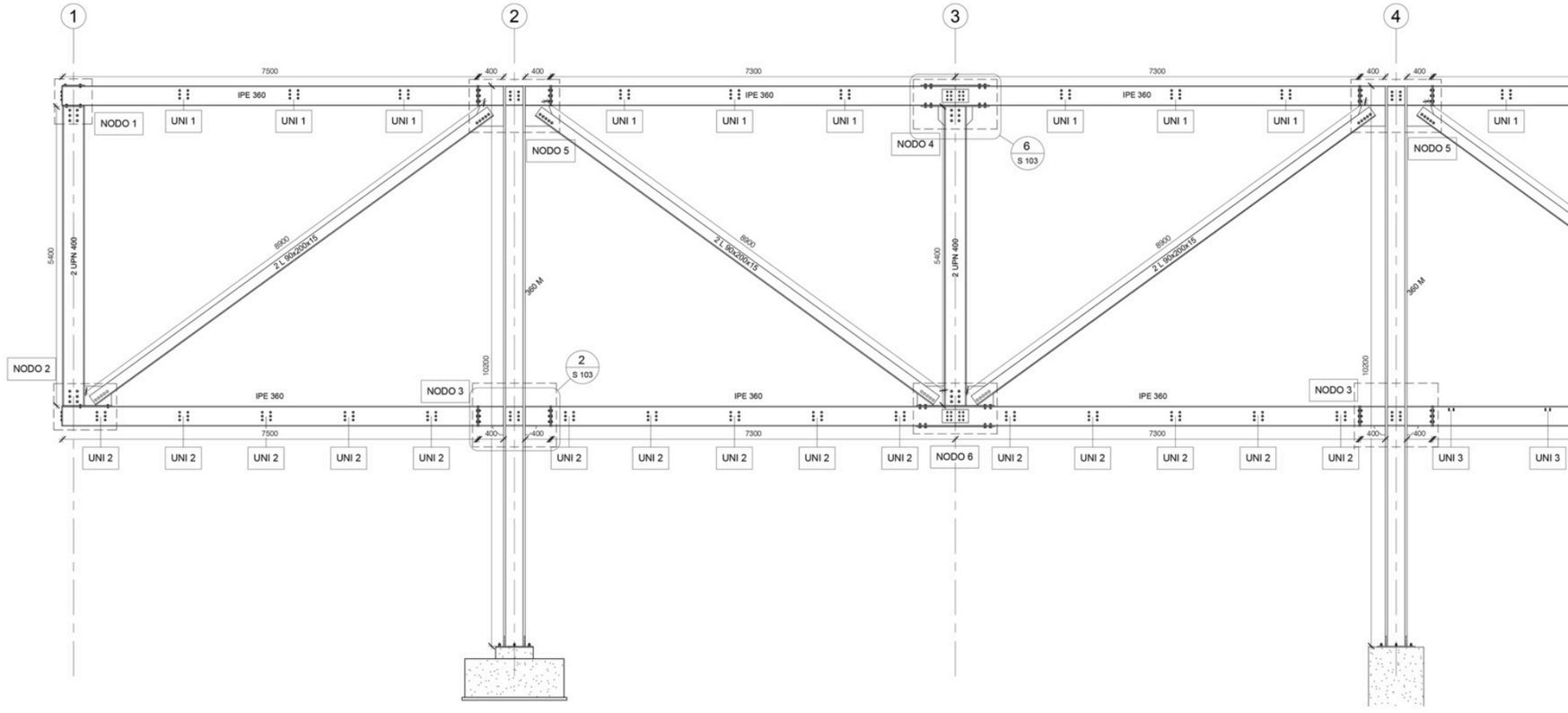


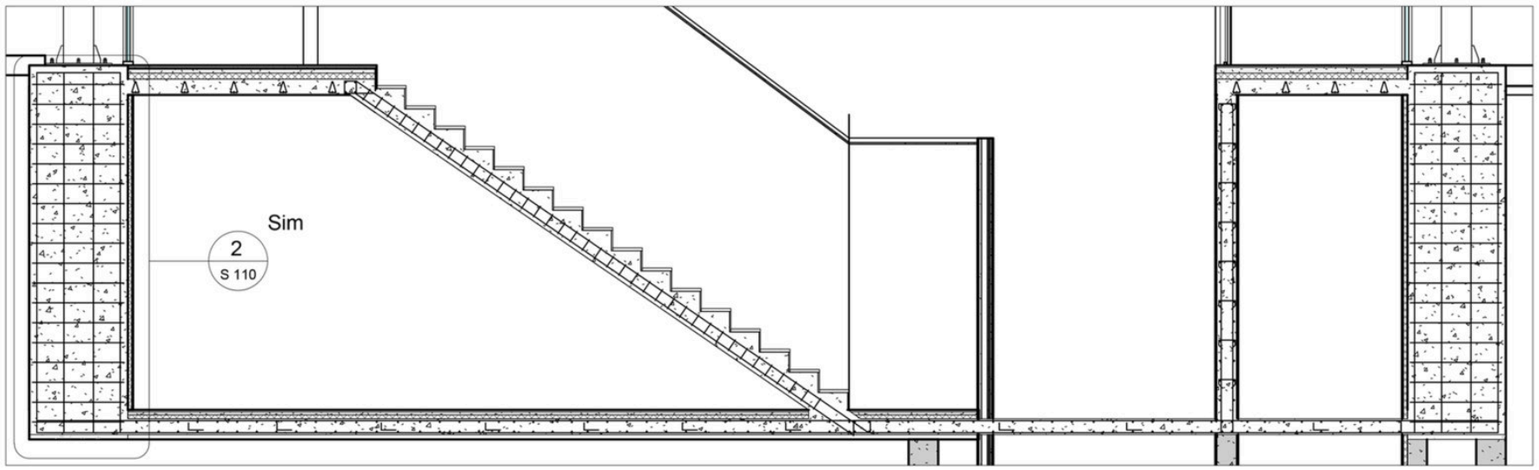

HEM 360
1378
piatto 350x1380
sp.20mm
2 L 90x200x15
IPE 360
piatto 170x1420 sp 15mm
con 8 bulloni 8.8 Ø24
1080

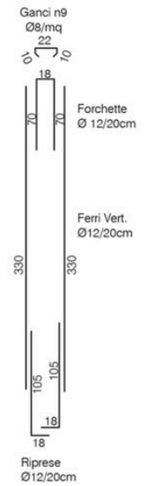

piatto 170x1420
sp 15mm con 8
bulloni 8.8 Ø24
HEA 340
unione bullonata con 2
squadrette 15x15x250mm
sp 8mm e bulloni Ø20 8.8
piatto 170x1420
sp 15mm con 8
bulloni 8.8 Ø24


2 L 90x200x15
2 UPN 400
2 L 90x200x15
IPE 360
HEA 340
IPE 360

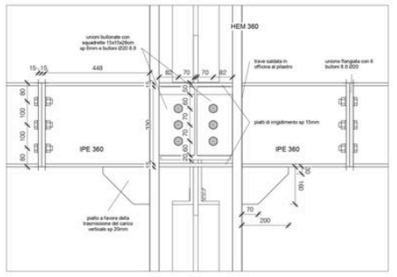

2 L 90x200x15
2 UPN 400
2 L 90x200x15
IPE 360
HEA 340

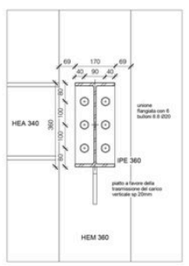

1
2
3
4
7500
400
400
7300
7300
IPE 360
NODO 1
UNI 1
NODO 5
NODO 4
6
S 103
5400
2 UPN 400
6900
2 L 90x200x15
360 M
NODO 2
10200
2
S 103
NODO 3
UNI 2
NODO 6
UNI 3

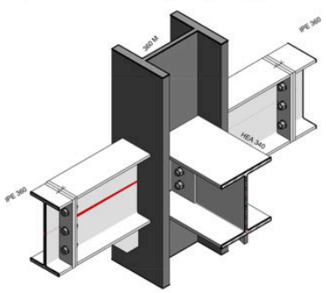

Sim
2
S 110

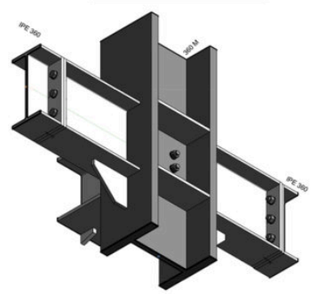

Ganci n9
Ø8/mq
22
10
10
18
70
70
Forchette
Ø 12/20cm
Ferri Vert.
Ø12/20cm
330
330
105
105
18
18
Riprese
Ø12/20cm

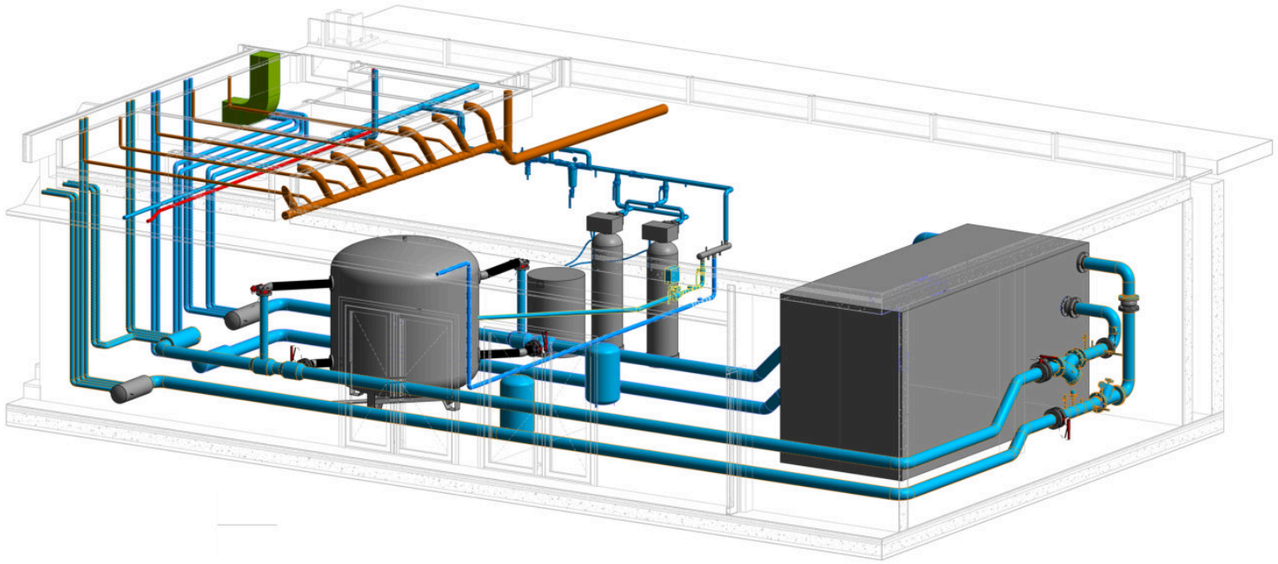

20
≤ 30
Distanziali 1Ø12/mq
70
70
Forchette Ø 12/20cm
18
700
Correnti Ø 12/20cm
700
Correnti Ø 12/20cm
70
70
Forchette Ø 12/20cm
18
700
Correnti Ø 12/20cm
700
Correnti Ø 12/20cm

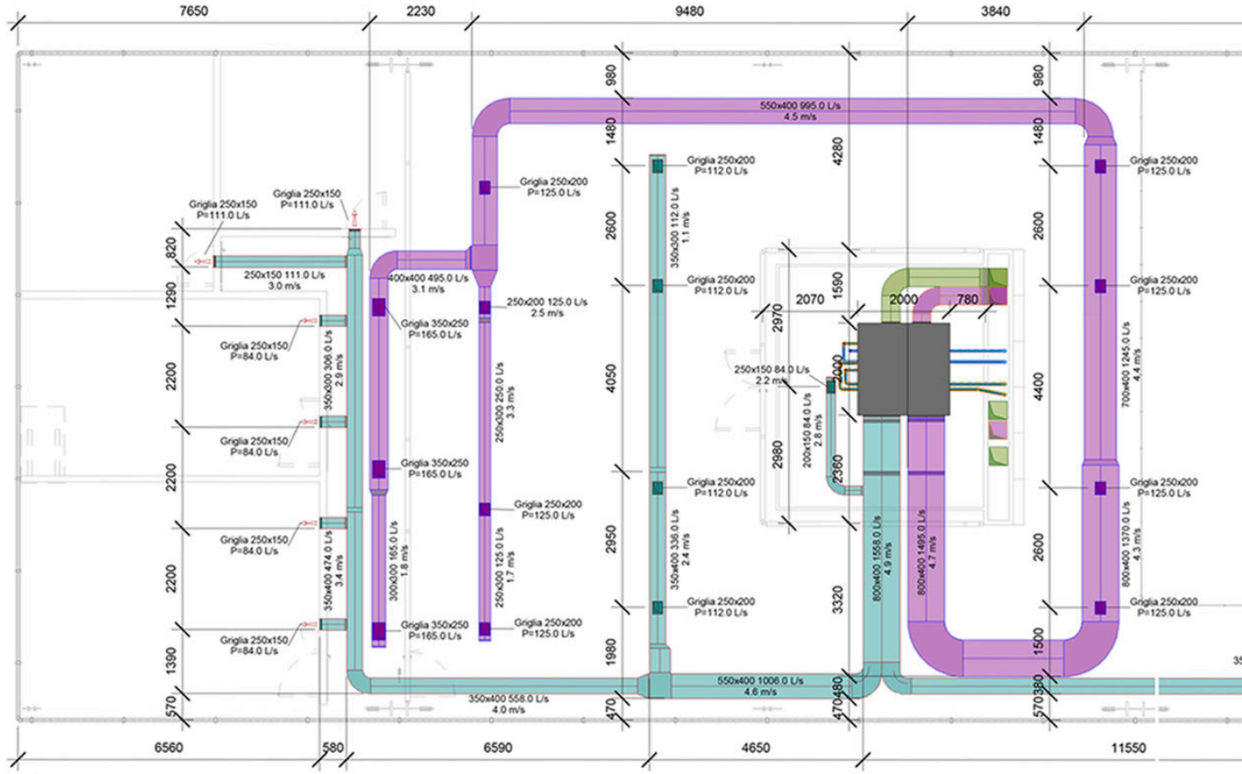

HEM 360
IPE 360
IPE 360
piatti di irrigidimento sp 15mm
piatto a favore della
trasmissione del carico
verticale sp 20mm

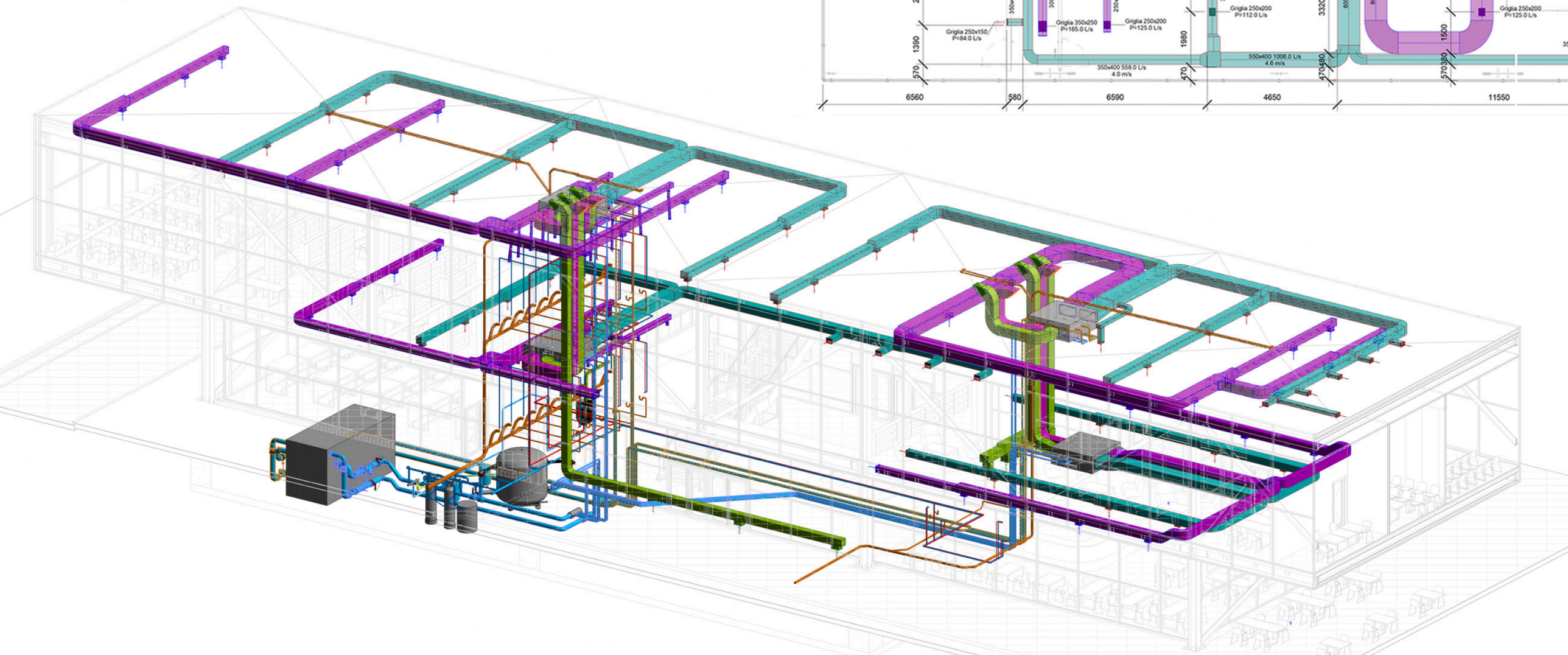

HEA 340
IPE 360
HEM 360


IPE 360
360 M
IPE 360
HEA 340
IPE 360


IPE 360
360 M
IPE 360




7650
2230
9480
3840
550x400 995.0 L/s
4.5 m/s
Griglia 250x150
P=111.0 L/s
Griglia 250x150
P=111.0 L/s
250x150 111.0 L/s
3.0 m/s
400x400 495.0 L/s
3.1 m/s
Griglia 250x200
P=125.0 L/s
Griglia 350x250
P=165.0 L/s
250x200 125.0 L/s
2.5 m/s
Griglia 250x150
P=84.0 L/s
350x300 306.0 L/s
2.9 m/s
250x300 250.0 L/s
3.3 m/s
Griglia 250x150
P=84.0 L/s
Griglia 350x250
P=165.0 L/s
Griglia 250x200
P=125.0 L/s
Griglia 250x150
P=84.0 L/s
350x400 474.0 L/s
3.4 m/s
300x300 165.0 L/s
1.8 m/s
250x300 125.0 L/s
1.7 m/s
Griglia 250x150
P=84.0 L/s
Griglia 350x250
P=165.0 L/s
Griglia 250x200
P=125.0 L/s
350x400 558.0 L/s
4.0 m/s
Griglia 250x200
P=112.0 L/s
350x300 112.0 L/s
1.1 m/s
Griglia 250x200
P=112.0 L/s
2070
1590
2000
780
250x150 84.0 L/s
2.2 m/s
200x150 84.0 L/s
2.8 m/s
Griglia 250x200
P=112.0 L/s
350x400 336.0 L/s
2.4 m/s
Griglia 250x200
P=112.0 L/s
800x400 1558.0 L/s
4.9 m/s
800x400 1495.0 L/s
4.7 m/s
550x400 1006.0 L/s
4.6 m/s
700x400 1245.0 L/s
4.4 m/s
Griglia 250x200
P=125.0 L/s
Griglia 250x200
P=125.0 L/s
Griglia 250x200
P=125.0 L/s
800x400 1370.0 L/s
4.3 m/s
Griglia 250x200
P=125.0 L/s
6560
580
6590
4650
11550



Chapter 03

Integra-AES; Studio Collaboration

Three commissions carried out with the design studio Integra-AES (Rome / La Spezia), spanning humanitarian, institutional and defence-sector clients; as Project Engineer and, on the third, as BIM Coordinator.

STUDIO

Integra-AES; Rome & La Spezia, Italy

ROLES

Project Engineer · BIM Coordinator

PERIOD

August 2023 – January 2024

SECTORS

Humanitarian · Institutional · Defence

The contents of these three commissions are under strict confidentiality; only a small number of images can be shown here. The MBDA Aulla drawing set in particular carries an explicit controlled-distribution marking.

AUG – NOV 2023 · PROJECT ENGINEER

WFP Brindisi

Collaborated with the team on the initial concept for the new main operational headquarters of the World Food Programme; accommodation buildings, research laboratories, a training area and a meeting area across a site of several hectares.

SEP – DEC 2023 · PROJECT ENGINEER

UNOPS Haiti

Coordinated the design of two buildings in two different locations, used as offices and medicine warehouses, through to delivery to the client. A hybrid reinforced-concrete and metal-carpentry structure was designed to resist earthquakes and hurricanes.

OCT 2023 – JAN 2024 · BIM COORDINATOR

MBDA Aulla

Coordinated to completion the design of several buildings intended as warehouses for explosive or military material, at a facility in Aulla (MS), Italy. Every part of the design was driven by protection indices and risk-reduction choices.

TECHNICAL NOTES; ENGINEERING SCOPE

STRUCTURAL APPROACH

UNOPS Haiti; hybrid reinforced-concrete and steel-carpentry structure, engineered for seismic and hurricane wind loads.

RISK-DRIVEN DESIGN

MBDA Aulla; building layout and separations calibrated against protection indices.

SPECIAL SYSTEMS

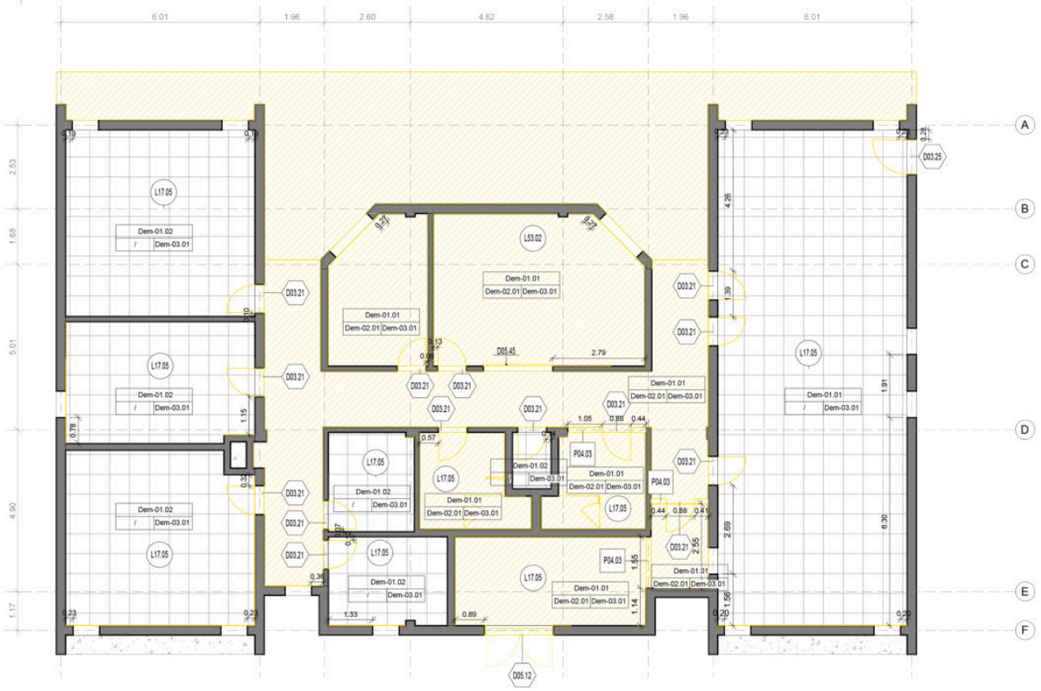
MBDA Aulla; design coordinated with the requirements of special machinery in environments containing explosive material, safety systems, and special air-treatment systems.

PROGRAMME SCALE

WFP Brindisi; masterplan-level concept across several hectares.

BIM COORDINATION

MBDA Aulla; BIM Coordinator role, from model management through to completion.



DR_Pianta Piano Terra Demolizioni - Groundfloor Plan Demolition
1 : 100

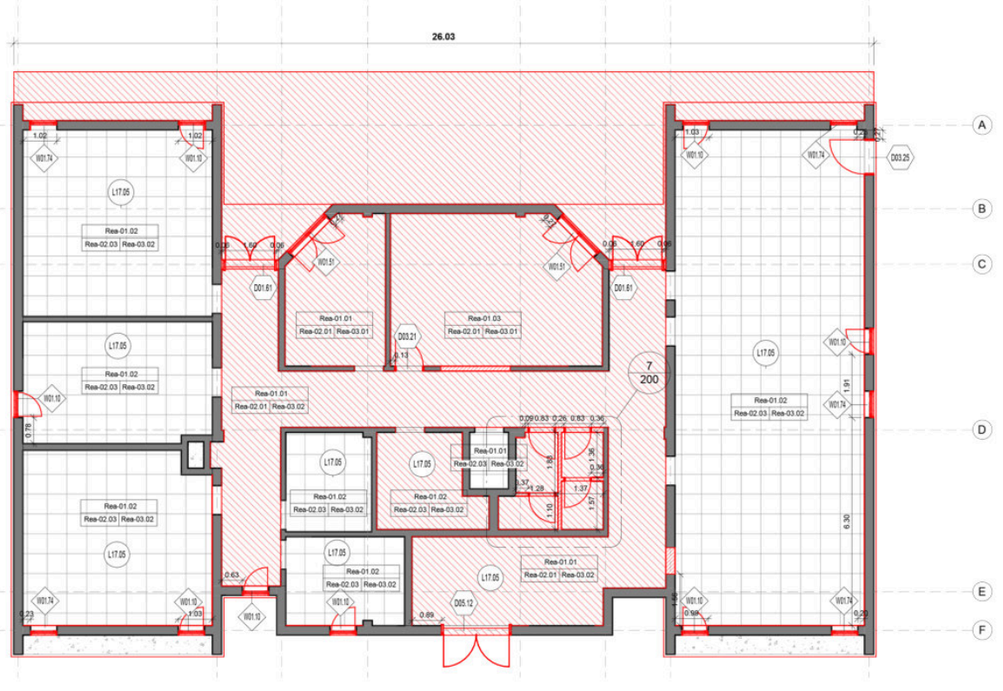
Elenco Lavorazioni in fase di Demolizione

Wall Finish		
Floor Finish		Ceiling Finish

Dem-01.01 Rimozione rivestimento setti e battiscopa
Dem-01.02 Rimozione battiscopa

Dem-02.01 Rimozione massetto e pavimentazione
Dem-02.02 Rimozione massetto pendenze e guaina impermeabile

Dem-03.01 Rimozione controsoffitto



DR_Pianta Piano Terra da Realizzare - Groundfloor Plan to Realize
1 : 100

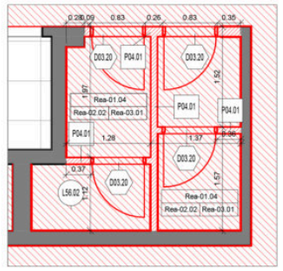
Elenco Lavorazioni da Realizzare

Wall Finish		
Floor Finish		Ceiling Finish

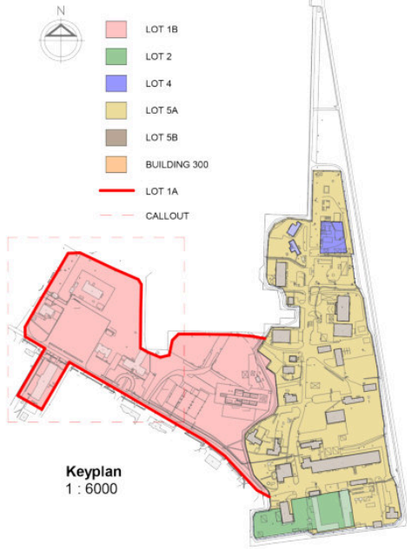
Rea-01.01 Consolidare intonaco distaccato e applicare una mano di tintura puntuale
Rea-01.02 Consolidare intonaco distaccato e applicare una mano di tintura nei punti di distacco e nella fascia alta scoperta dalla rimozione del controsoffitto
Rea-01.03 Consolidare intonaco e ritinteggiare interamente
Rea-01.04 Applicare uno strato di malta per la presa di piastrelle ceramiche e inserire un battiscopa waterproof

Rea-02.01 Realizzare massetto e strato a finitura in vinile
Rea-02.02 Realizzare massetto e strato a finitura in piastrelle ceramiche
Rea-02.03 Pulire e recuperare dove possibile pavimentazione esistente
Rea-02.04 Realizzare massetto delle pendenze e guaina impermeabile

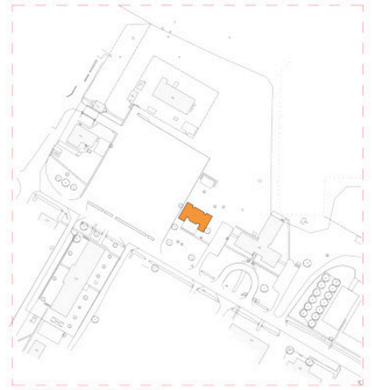
Rea-03.01 Realizzare controsoffitto modulare 60x60
Rea-03.02 Intonacare soffitto



Focus Lavorazione Area Bagni / Callout of Bathroom Area
1 : 50

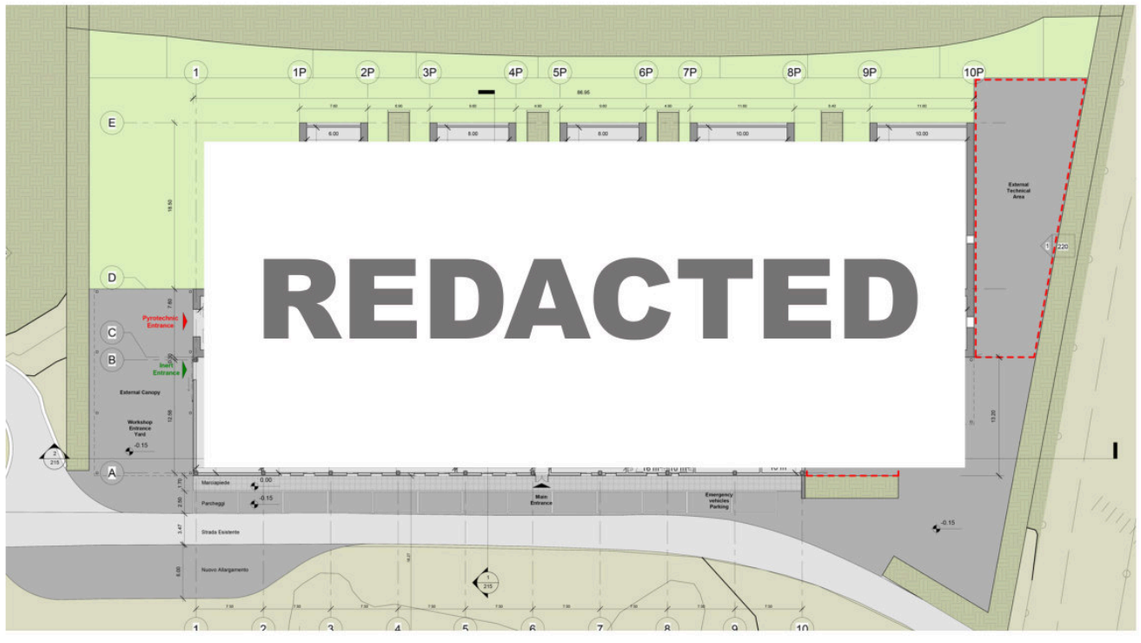


Keyplan
1 : 6000



LOT 1B_Building 300
1 : 3000





Chapter 04

Verona Villafranca Airport

Co-head of the construction site technical office for the expansion and refurbishment of the departures terminal at Verona's Aeroporto Valerio Catullo, with construction company Itinera S.p.A.; bringing the executive project, including the entire civil scope, to construction-site level.

LOCATION
Verona, Italy

CONSTRUCTION COMPANY
Itinera S.p.A. (ASTM Group)

ROLE
Co-Head, construction site technical office

PERIOD
January – June 2024



New departures terminal — exterior

TECHNICAL NOTES; ENGINEERING SCOPE

SITE ENGINEERING

Adapted the executive design to as-built conditions and client/supplier requirements; managed quality control and compliance checks.

STRUCTURAL SYSTEM

Composite steel-concrete floors; HEA/HEB/IPE beams (HEA600, HEB700, IPE360/400/450) with Hilti X-HVB95 shear studs on trapezoidal decking and a 12 cm RC topping slab.

RETROFIT & STRENGTHENING

Existing steel beams reinforced with bolted S355 side plates (M20 cl. 8.8); foundation buckets, column sections and precast beam joints strengthened with chemically-anchored rebar (Hilti HY 200-A resin, M16/M20).

ENVELOPE

Full-height glazed curtain-wall facade.

MEP COORDINATION

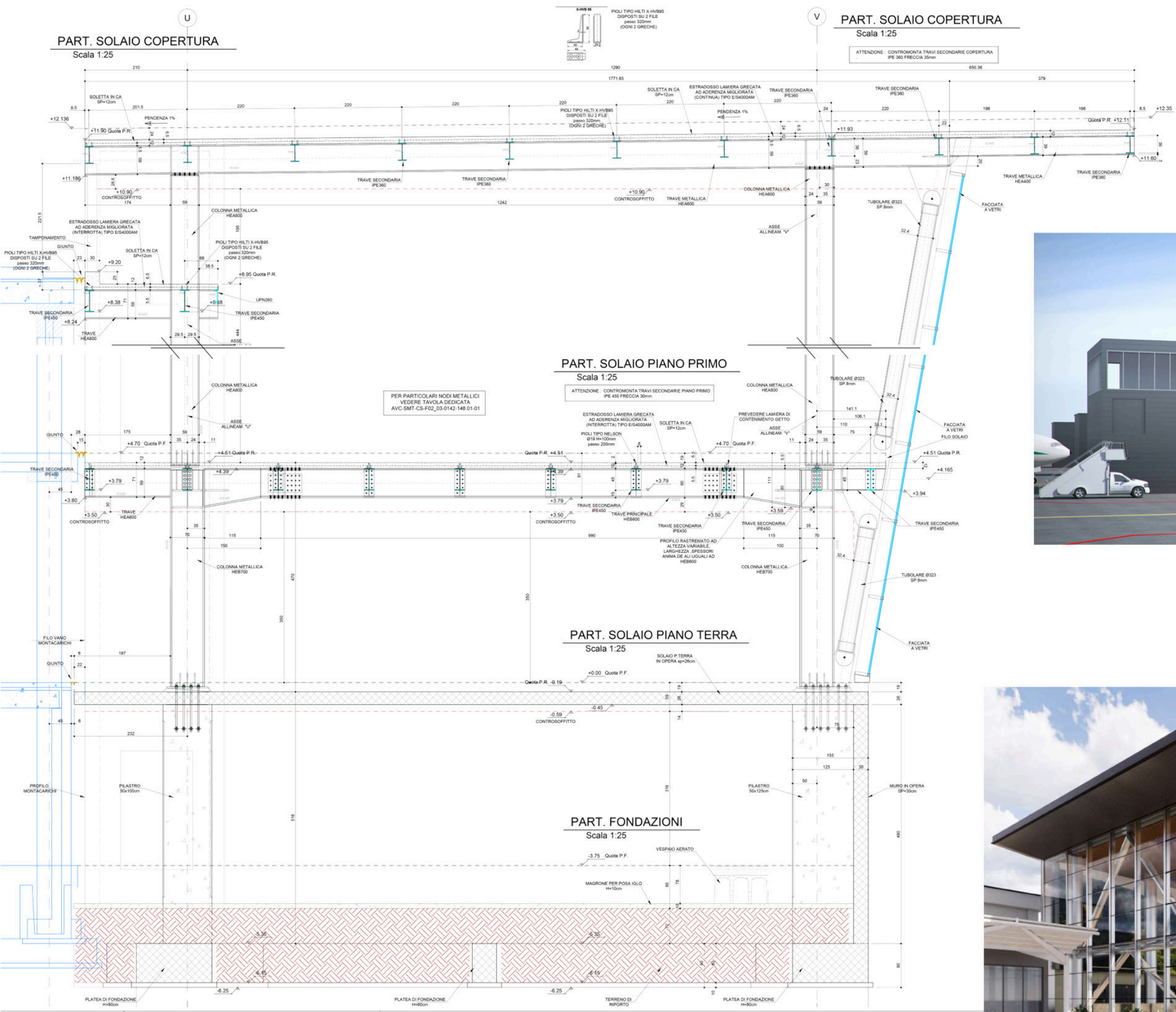
Design coordination with MEP (mechanical, electrical and plumbing) requirements.

PROCUREMENT

Managed procurement for windows/doors, lifts, escalators and ceiling finishes, per updated requirements.

VARIATION MANAGEMENT

Managed the cost side of all necessary project variations with site management and the client.


U
PART. SOLAIO COPERTURA
Scala 1:25
X-HVB 95
PIOLI TIPO HILTI X-HVB95
DISPOSTI SU 2 FILE
passo 320mm
(OGNI 2 GRECHE)
V
PART. SOLAIO COPERTURA
Scala 1:25
ATTENZIONE : CONTROMONTA TRAVI SECONDARIE COPERTURA
IPE 360 FRECCIA 35mm
SOLETTA IN CA SP=12cm
PENDENZA 1%
ESTRADOSSO LAMIERA GRECATA AD ADERENZA MIGLIORATA (CONTINUA) TIPO E/S4000AM
TRAVE SECONDARIA IPE360
TRAVE METALLICA HEA600
COLONNA METALLICA HEA600
TUBOLARE Ø323 SP.8mm
FACCIATA A VETRI
ASSE ALLINEAM. "V"
ESTRADOSSO LAMIERA GRECATA AD ADERENZA MIGLIORATA (INTERROTTA) TIPO E/S4000AM
TAMPONAMENTO
GIUNTO
SOLETTA IN CA SP=12cm
TRAVE SECONDARIA IPE450
TRAVE HEA600
UPN280
ASSE
PART. SOLAIO PIANO PRIMO
Scala 1:25
ATTENZIONE : CONTROMONTA TRAVI SECONDARIE PIANO PRIMO
IPE 450 FRECCIA 30mm
PER PARTICOLARI NODI METALLICI
VEDERE TAVOLA DEDICATA
AVC-SMT-CS-F02_03-0142-148.01-01
PREVEDERE LAMIERA DI CONTENIMENTO GETTO
PIOLI TIPO NELSON Ø19 H=100mm passo 200mm
TRAVE PRINCIPALE HEB600
PROFILO RASTREMATO AD ALTEZZA VARIABILE, LARGHEZZA ,SPESSORI ANIMA DE ALI UGUALI AD HEB600
COLONNA METALLICA HEB700
FILO SOLAIO
PART. SOLAIO PIANO TERRA
Scala 1:25
SOLAIO P.TERRA IN OPERA sp=26cm
FILO VANO MONTACARICHI
PROFILO MONTACARICHI
PILASTRO 50x100cm
PILASTRO 50x125cm
MURO IN OPERA SP=30cm
PART. FONDAZIONI
Scala 1:25
VESPAIO AERATO
MAGRONE PER POSA IGLO H=10cm
PLATEA DI FONDAZIONE H=80cm
TERRENO DI RIPORTO





Chapter 05

Venice Marco Polo Airport

Head of the construction site technical office, BIM Manager and BIM Coordinator for the new terminal at Venice's Marco Polo Airport, with construction company Itinera S.p.A.; coordinating the full design process through to delivery on site.

LOCATION

Venice, Italy

CONSTRUCTION COMPANY

Itinera S.p.A. (ASTM Group)

ROLE

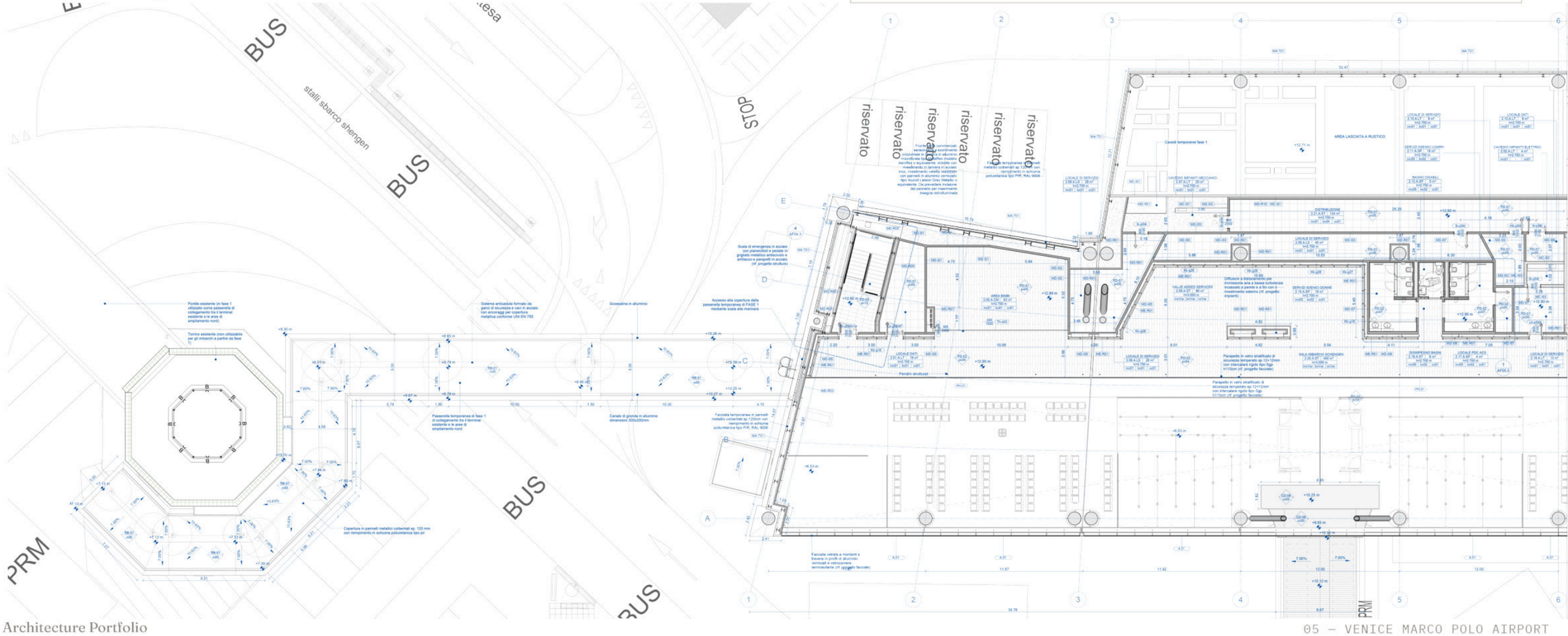
Head, technical office · BIM Manager & Coordinator

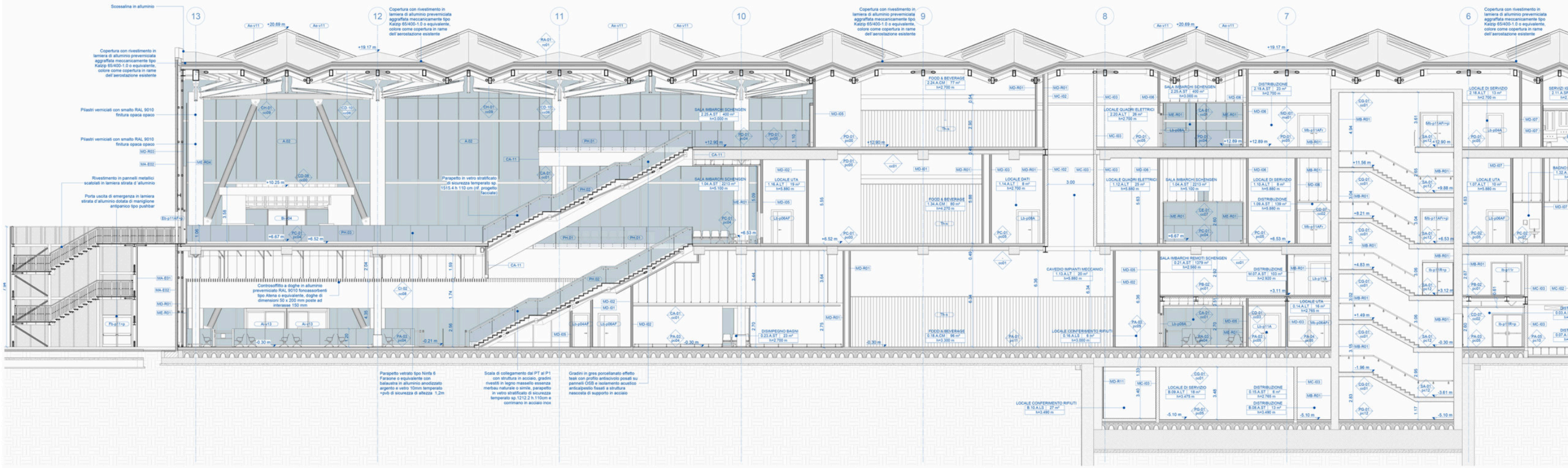
PERIOD

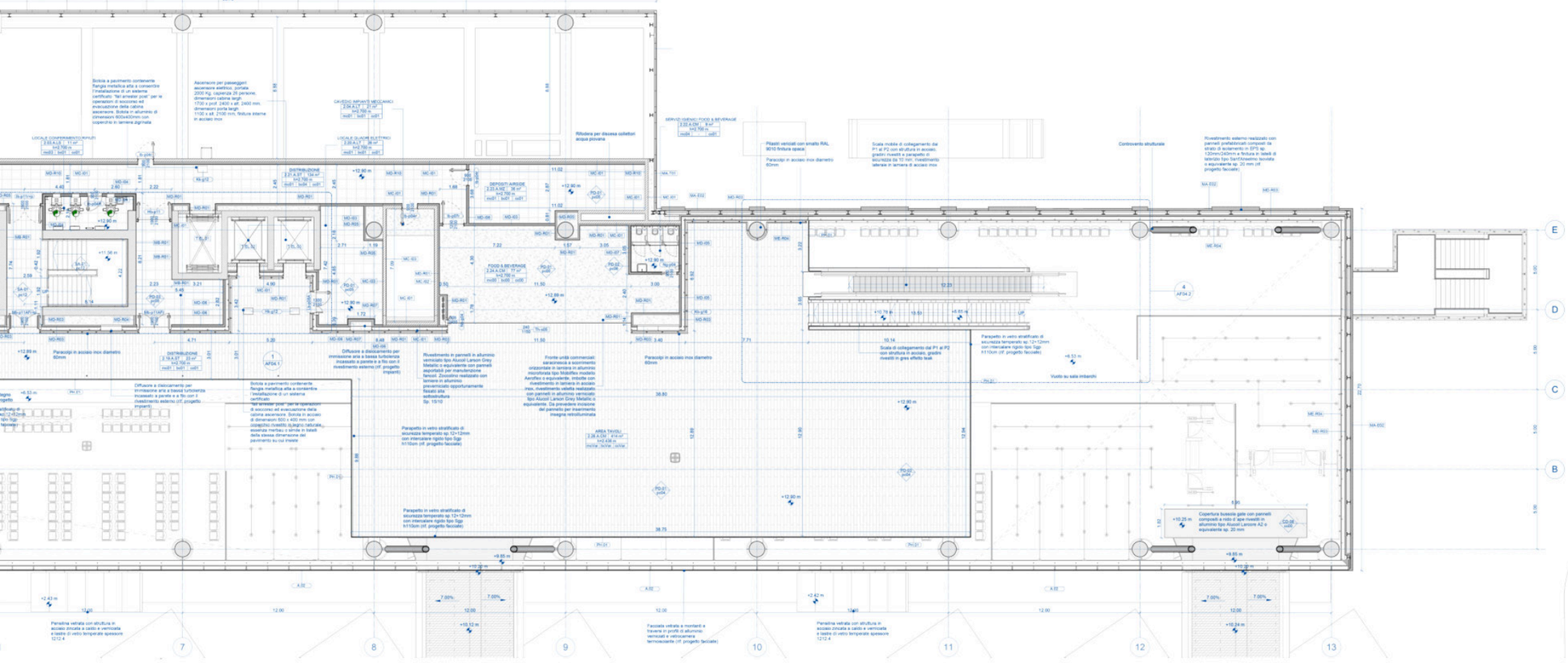
June – December 2024



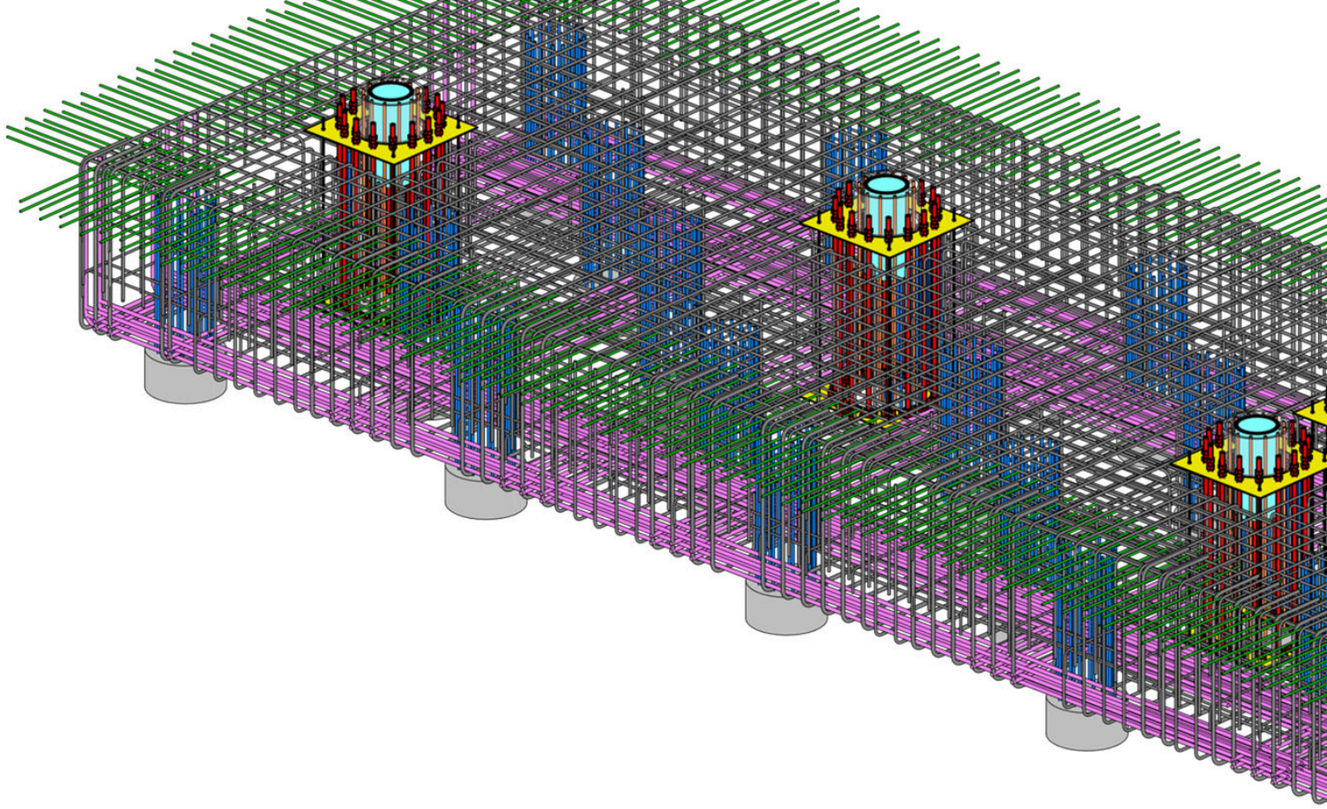
New terminal — apron side




Scossalina in alluminio
Copertura con rivestimento in lamiera di alluminio preverniciata aggraffata meccanicamente tipo Kalzip 65/400-1.0 o equivalente, colore come copertura in rame dell'aerostazione esistente
Pilastri verniciati con smalto RAL 9010 finitura opaca opaco
Rivestimento in pannelli metallici scatolati in lamiera stirata d'alluminio
Porta uscita di emergenza in lamiera stirata d'alluminio dotata di maniglione antipanico tipo pushbar
Parapetto in vetro stratificato di sicurezza temperato sp. 1515.4 h 110 cm (rif. progetto facciate)
Controsoffitto a doghe in alluminio preverniciato RAL 9010 fonoassorbenti tipo Atena o equivalente, doghe di dimensioni 50 x 200 mm poste ad interasse 150 mm
SALA IMBARCHI SCHENGEN
LOCALE UTA
FOOD & BEVERAGE
LOCALE DATI
CAVEDIO IMPIANTI MECCANICI
DISIMPEGNO BAGNI
LOCALE CONFERIMENTO RIFIUTI
LOCALE QUADRI ELETTRICI
SALA IMBARCHI REMOTI SCHENGEN
DISTRIBUZIONE
LOCALE DI SERVIZIO
Parapetto vetrato tipo Ninfa 6 Faraone o equivalente con balaustra in alluminio anodizzato argento e vetro 10mm temperato +pvb di sicurezza di altezza 1,2m
Scala di collegamento dal PT al P1 con struttura in acciaio, gradini rivestiti in legno massello essenza merbau naturale o simile, parapetto in vetro stratificato di sicurezza temperato sp.1212.2 h.110cm e corrimano in acciaio inox
Gradini in gres porcellanato effetto teak con profilo antiscivolo posati su pannelli OSB e isolamento acustico anticalpestio fissati a struttura nascosta di supporto in acciaio


Bolcia a pavimento contenente flangia metallica atta a consentire l'installazione di un sistema certificato "air breather pipe" per le operazioni di soccorso ed evacuazione della cabina ascensore. Bolcia in alluminio di dimensioni 600x400mm con coperchio in lamiera zigrinata
Ascensore per passeggeri, ascensore elettrico, portata 2000 Kg, capienza 26 persone, dimensioni cabina: largh. 1730 x prof. 2400 x alt. 2400 mm, dimensioni porta: largh. 1100 x alt. 2100 mm, finiture interne in acciaio inox
CAVEDIO IMPIANTO MECCANICO
LOCALE QUADRI ELETTRICI
DISTRIBUZIONE
Ribadera per discesa collettori acqua piovana
SERVIZI IGIENICI FOOD & BEVERAGE
DEPOSITI AIRSIDE
FOOD & BEVERAGE
Pilastri verniciati con smalto RAL 9010 finitura opaca
Parapetto in acciaio inox diametro 40mm
Scala mobile di collegamento dal P1 al P2 con struttura in acciaio, gradini rivestiti a parapetto di sicurezza da 10 mm. Investimento laterale in lamiera di acciaio inox
Controvento strutturale
Rivestimento esterno realizzato con pannelli prefabbricati composti da strato di isolamento in EPS sp. 120mm/240mm e finitura in laterit di laterizio tipo Sant'Antonino Isovista o equivalente sp. 20 mm (rif. progetto facciate)
Paraspigoli in acciaio inox diametro 60mm
Diffusore a disco/ugello per immissione aria a bassa turbolenza incassato a parete e a filo con il rivestimento esterno (rif. progetto impianti)
Rivestimento in pannelli in alluminio verniciato tipo Alucoil Larson Grey Metallic o equivalente con pannelli asportabili per manutenzione fancoil. Zoccolino realizzato con lamiere in alluminio preverniciato opportunamente fissato alla sottostruttura Sp. 10/10
Fronte unità commerciali: paraschiena e scorrimento orizzontale in lamiera in alluminio microforata tipo Modoffice modello Aerofilec o equivalente, imbotte con rivestimento in lamiera in acciaio inox, rivestimento velatella realizzato con pannelli in alluminio verniciato tipo Alucoil Larson Grey Metallic o equivalente. Da prevedere incisione del pannello per inserimento insegna retroilluminata
Parapetto in vetro stratificato di sicurezza temperato sp. 12+12mm con interlacciare rigido tipo Sgp h110cm (rif. progetto facciate)
Vuoto su sala imbarchi
Copertura bussola gate con pannelli composti a nido d'ape rivestiti in alluminio tipo Alucoil Larsore A2 o equivalente sp. 20 mm
Facciata vetrata a montanti e traversi in profili di alluminio verniciati e vetrocamera termoisolante (rif. progetto facciate)
Pensilina vetrata con struttura in acciaio zincata a caldo e verniciata e lastre di vetro temperate spessore 12/12.4





TECHNICAL NOTES; ENGINEERING SCOPE

SITE ENGINEERING

Adapted the executive design to as-built conditions and client/supplier requirements; managed quality control and compliance checks.

STRUCTURAL SYSTEM

Hybrid precast structure, with reinforced steel profiles subsequently encased in cast concrete, on pad foundations with deep piles or diaphragm walls.

BIM SCOPE

Information Management Plan, Asset Information Requirements and Common Data Environment (CDE) management across the architectural, structural and MEP models.

PROCUREMENT

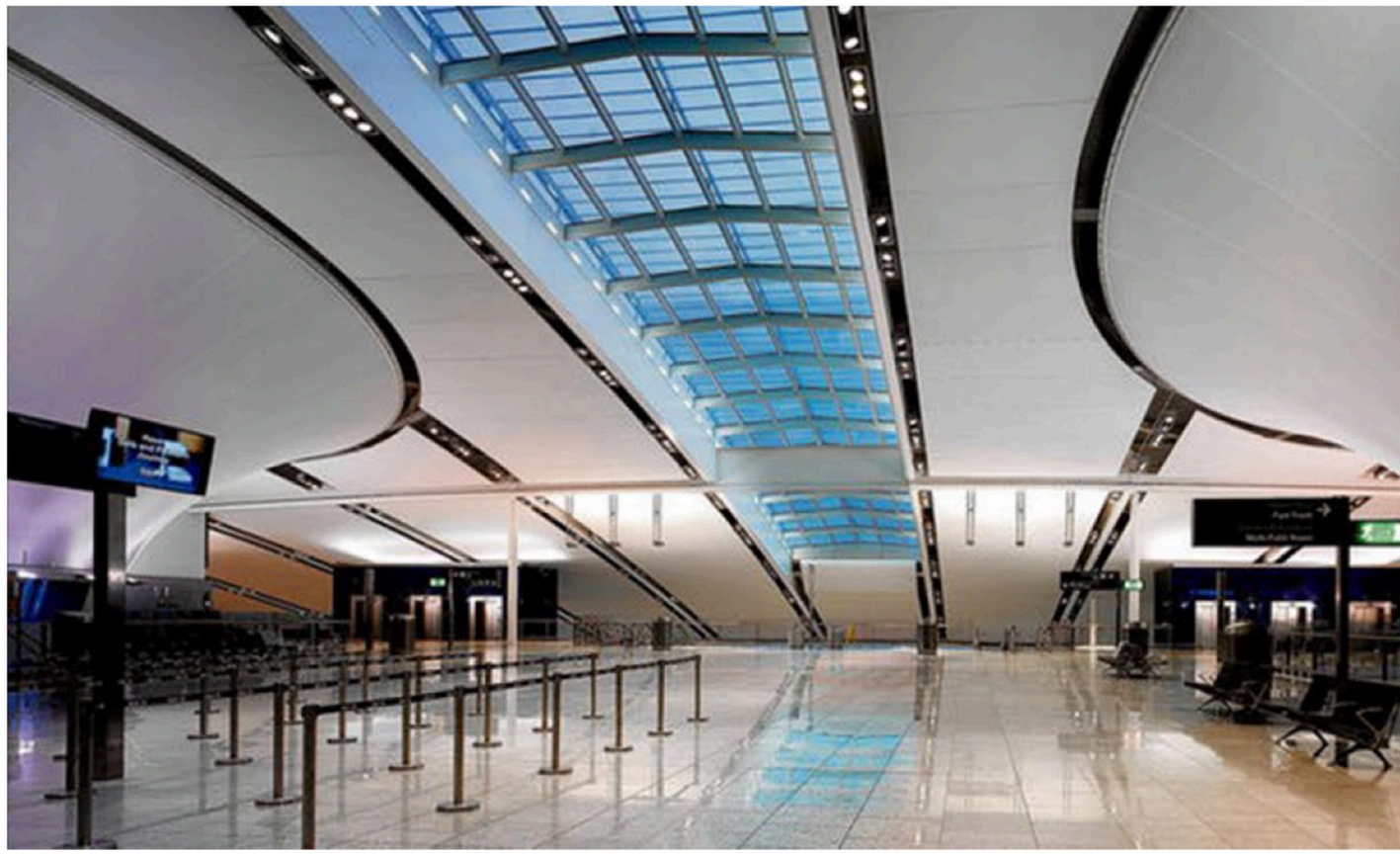
Managed procurement for the technical steel structure system, predalles, lifts and escalators, waterproofing systems and the glazed facade, per updated requirements.

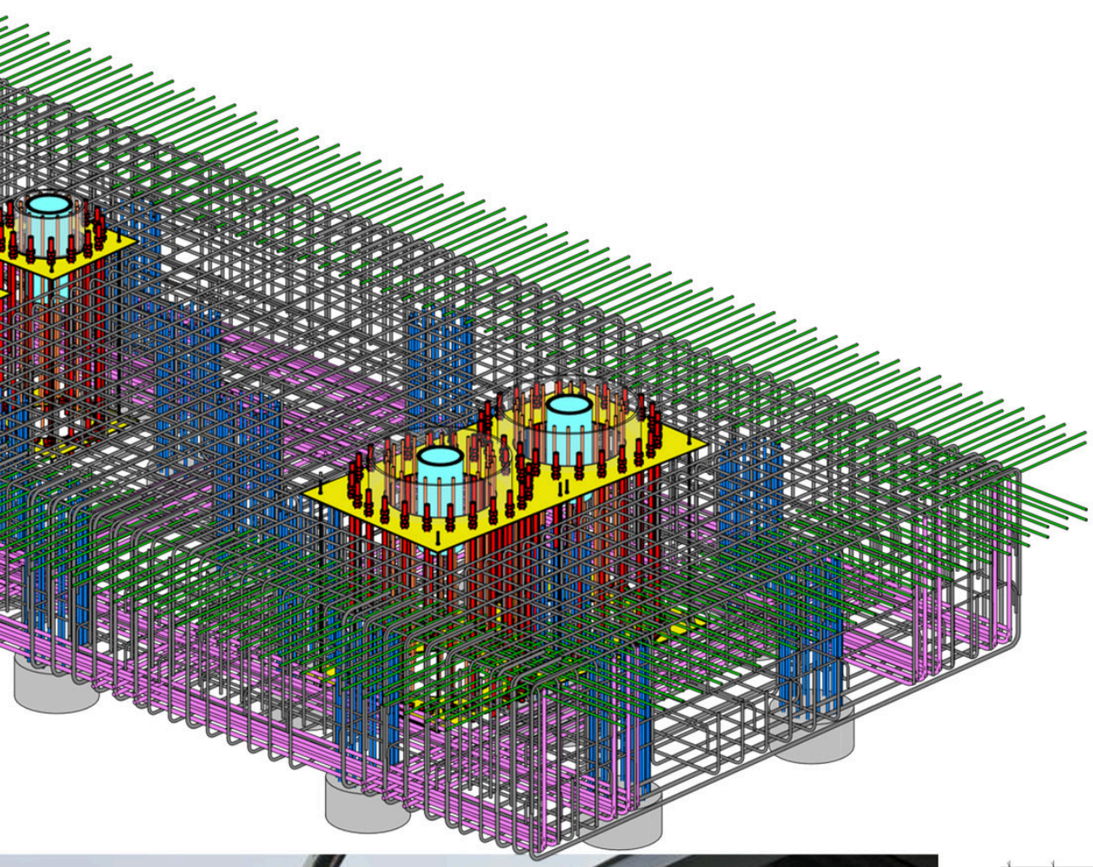
DESIGN STUDIO COORDINATION

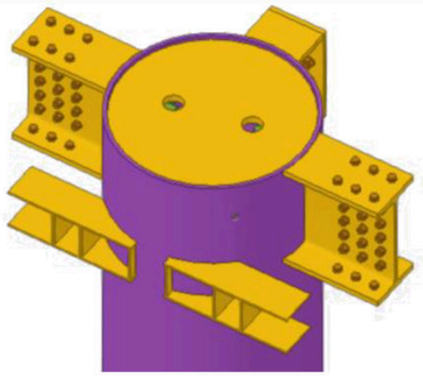
Managed external design teams for the redesign of structural and architectural elements, coordinating with MEP, suppliers, and site requirements and schedules.

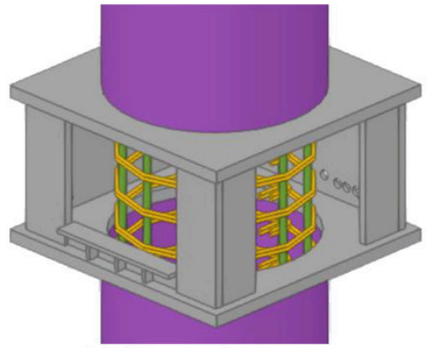
SITE APPROVALS

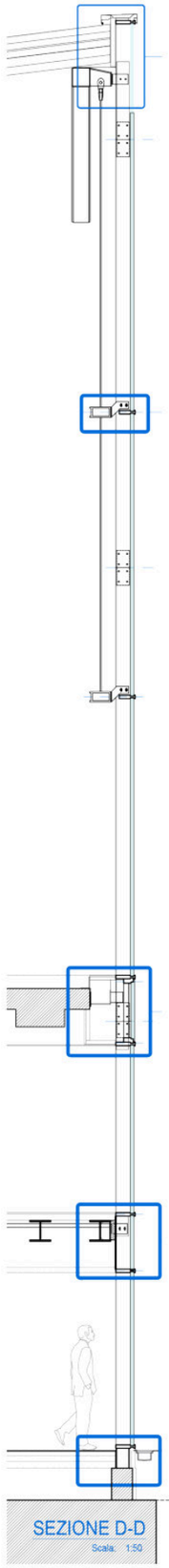
Developed and managed the approval of construction documents: drawings, schedule, material data sheets, load test reports and BIM documents.

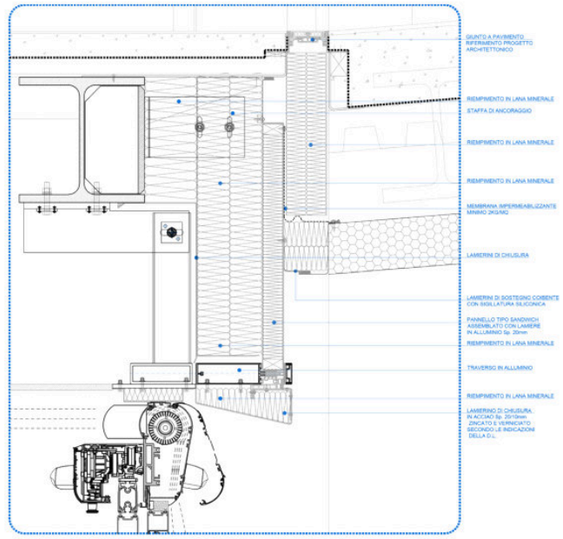


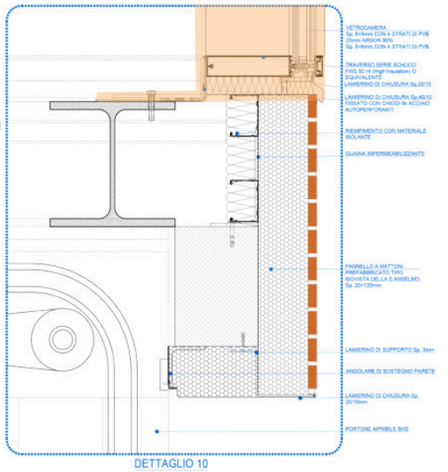




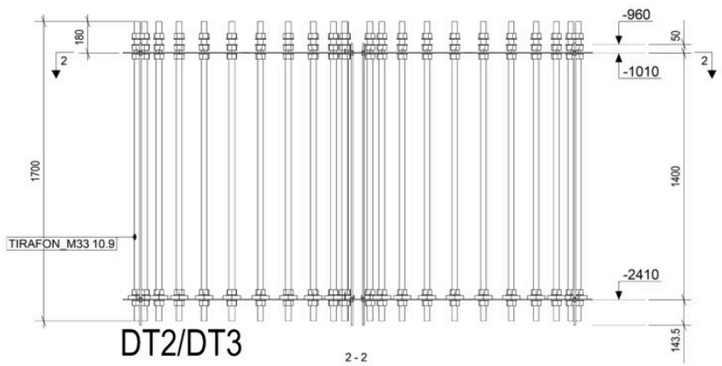




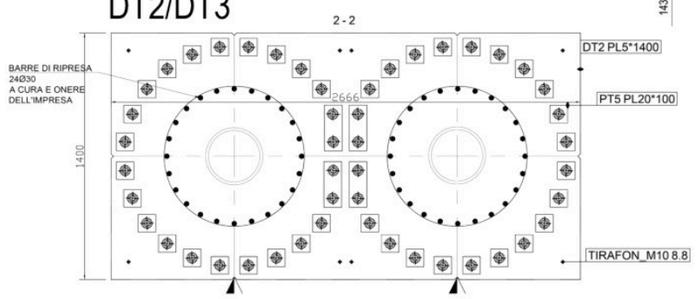


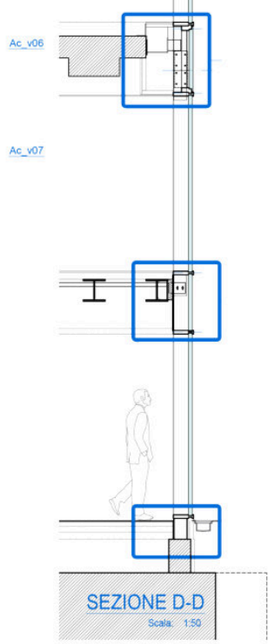


DETTAGLIO 10
Scala 1:5



DT2/DT3





SEZIONE D-D
Scala 1:50

Chapter 06

The White Lotus Koh Samui

A twenty-unit residential condominium on Koh Samui, carried from an aesthetic concept, developed in parallel with its structural framework, through to a model detailed enough for an initial construction cost estimate. The flagship project of Kolibri Studio, the practice I co-founded in June 2024.

TYPOLOGY

Residential condominium; 20 units

ROLE

Co-Founder & Architectural Project Manager, Kolibri Studio

LOCATION

Koh Samui, Thailand

STATUS

Ongoing; June 2024 to present



Kolibri Studio logo

TECHNICAL NOTES; ENGINEERING SCOPE

STRUCTURAL FRAMEWORK

Shear-wall structure with ribbed slabs to support the large cantilevers and variable loads, on strap-beam foundations for maximum ground bearing.

WALL TYPES

GYP; gypsum board on metal studs · MAN; clay-block masonry · WOD; timber battens on vertical timber studs.

PROCUREMENT

Selection of key suppliers to achieve the required aesthetic and quality standard.

MEP SCOPE

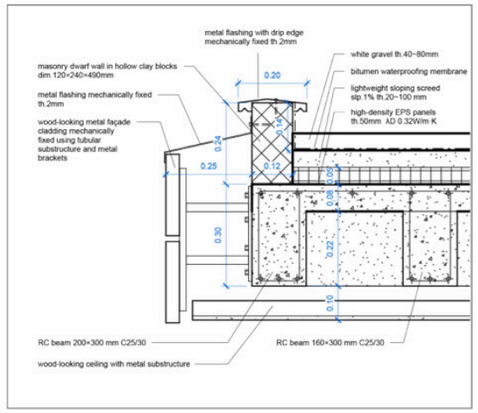
Full calculations for the mechanical and plumbing systems, including a dedicated irrigation system and a pool water-circulation system.

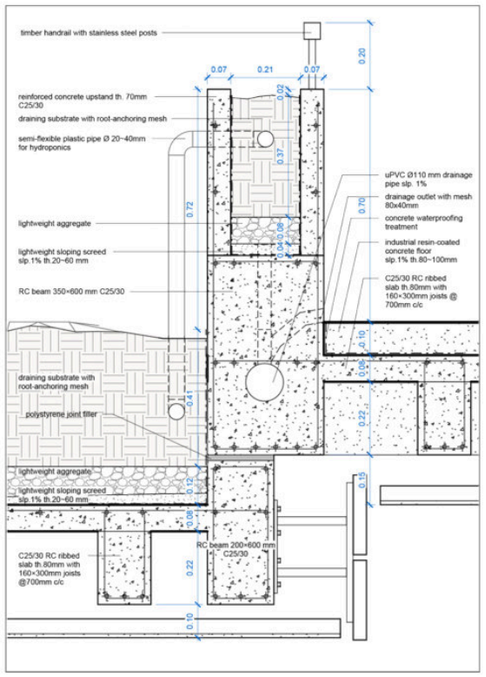
COST ESTIMATION

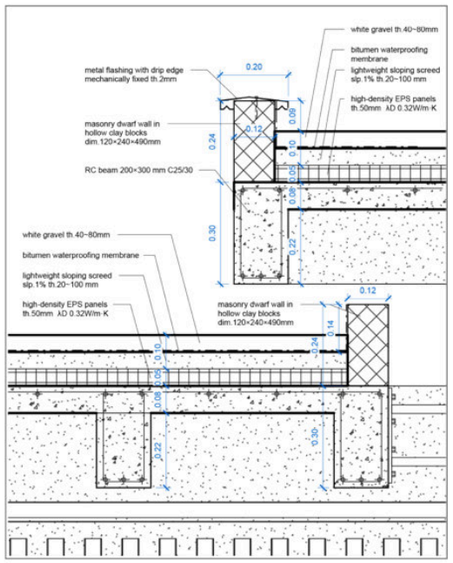
Quantity take-off and definition of construction and fit-out costs.

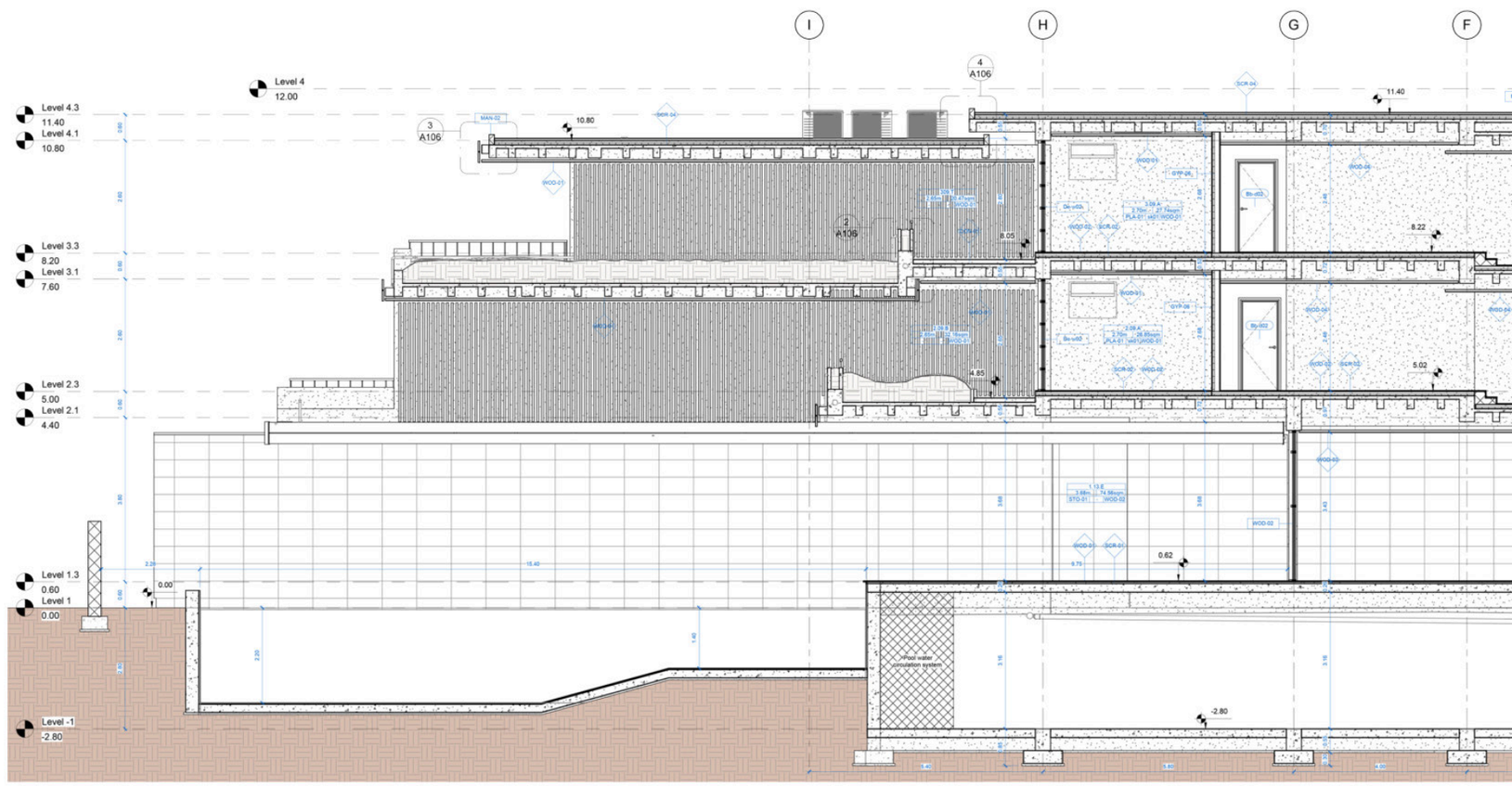
DOCUMENTATION

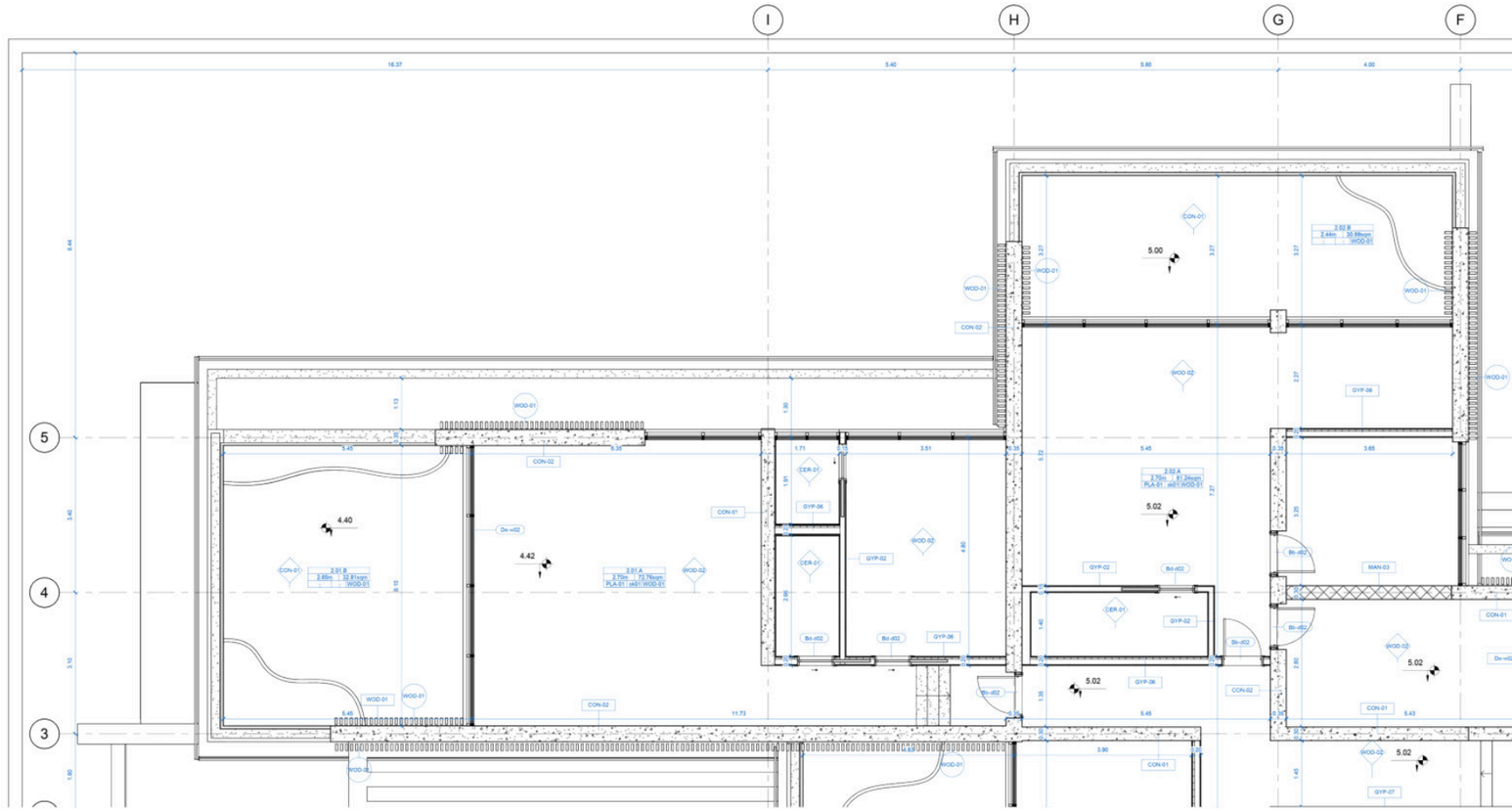
Development of project drawings and documentation for building permits.


metal flashing with drip edge
mechanically fixed th.2mm
masonry dwarf wall in hollow clay blocks
dim.120×240×490mm
metal flashing mechanically fixed
th.2mm
wood-looking metal façade
cladding mechanically
fixed using tubular
substructure and metal
brackets
white gravel th.40~80mm
bitumen waterproofing membrane
lightweight sloping screed
slp.1% th.20~100 mm
high-density EPS panels
th.50mm λD 0.32W/m·K
RC beam 200×300 mm C25/30
RC beam 160×300 mm C25/30
wood-looking ceiling with metal substructure


timber handrail with stainless steel posts
reinforced concrete upstand th. 70mm
C25/30
draining substrate with root-anchoring mesh
semi-flexible plastic pipe Ø 20~40mm
for hydroponics
lightweight aggregate
lightweight sloping screed
slp.1% th.20~60 mm
RC beam 350×600 mm C25/30
uPVC Ø110 mm drainage
pipe slp. 1%
drainage outlet with mesh
80x40mm
concrete waterproofing
treatment
industrial resin-coated
concrete floor
slp.1% th.80~100mm
C25/30 RC ribbed
slab th.80mm with
160×300mm joists @
700mm c/c
draining substrate with
root-anchoring mesh
polystyrene joint filler
lightweight aggregate
lightweight sloping screed
slp.1% th.20~60 mm
RC beam 200×600 mm
C25/30
C25/30 RC ribbed
slab th.80mm with
160×300mm joists
@700mm c/c


metal flashing with drip edge
mechanically fixed th.2mm
masonry dwarf wall in
hollow clay blocks
dim.120×240×490mm
RC beam 200×300 mm C25/30
white gravel th.40~80mm
bitumen waterproofing membrane
lightweight sloping screed
slp.1% th.20~100 mm
high-density EPS panels
th.50mm λD 0.32W/m·K
masonry dwarf wall in
hollow clay blocks
dim.120×240×490mm


I
H
G
F
Level 4
12.00
Level 4.3
11.40
Level 4.1
10.80
Level 3.3
8.20
Level 3.1
7.60
Level 2.3
5.00
Level 2.1
4.40
Level 1.3
0.60
Level 1
0.00
Level -1
-2.80
11.40
10.80
8.22
8.05
5.02
4.85
0.62
0.00
-2.80
Pool water circulation system


I
H
G
F
5
4
3
5.00
4.40
4.42
5.02



ORGANICITY

Tropical Harmony

The White Lotus is conceived as an organic extension of Koh Samui's landscape rather than an imposition on it. Following the language of Frank Lloyd Wright, vertical structural walls and glazing meet the horizontal lines of floor bands and cantilevered balconies, softened by a green veil of planting that runs across every level and into the surrounding garden. Stone and timber cladding, and a planting palette ranging from lush native species to minimalist Japanese-inspired greenery, let the architecture shift with the seasons; four closed-loop hydroponic systems keep it self-sufficient with almost no maintenance.

ELEGANCE

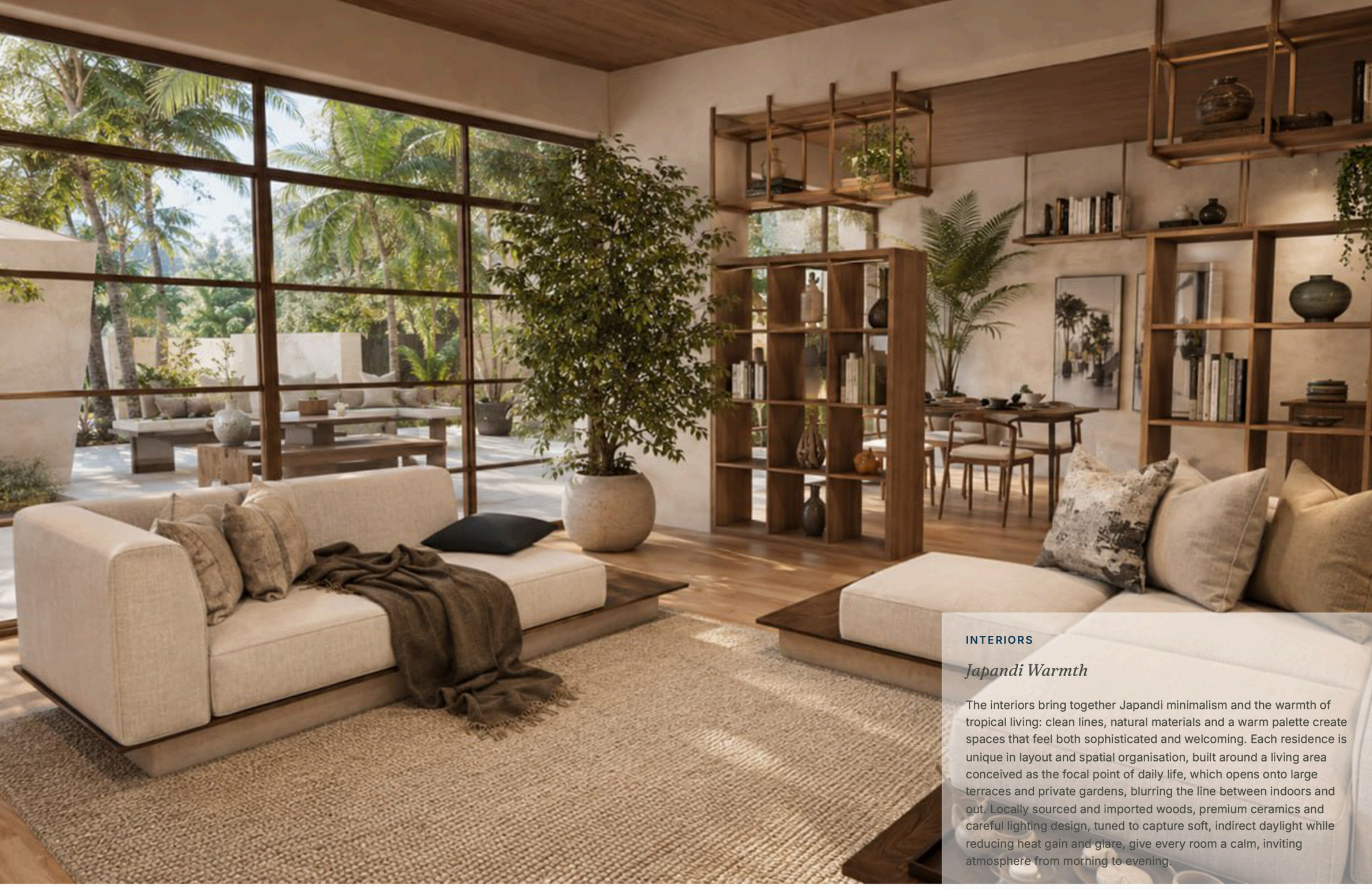
Simplicity and Luxury

Elegance here is not ostentation but balance: noble materials, from natural stone to warm timber and refined metal, are combined for both their beauty and their durability. Clean lines, balanced volumes and an absence of anything superfluous give the residence a calm, considered character. Every proportion, from the facades to the common areas, is studied so the building reads as one coherent, fluid whole: a place designed to be lived in with pride, today and for years to come.

SUSTAINABILITY

Green Living & Smart Technology

Sustainability is treated as a design discipline in its own right, not a trade-off against comfort. An advanced photovoltaic system with battery storage cuts dependence on the grid, while harvested rainwater feeds the hydroponic irrigation of the green spaces. Orientation, cross-ventilation and high-performance, low-impact insulation respond to the local climate, and the site's existing greenery is preserved and woven into the new landscape, so the project regenerates its surroundings as much as it inhabits them.



INTERIORS

Japandi Warmth

The interiors bring together Japandi minimalism and the warmth of tropical living: clean lines, natural materials and a warm palette create spaces that feel both sophisticated and welcoming. Each residence is unique in layout and spatial organisation, built around a living area conceived as the focal point of daily life, which opens onto large terraces and private gardens, blurring the line between indoors and out. Locally sourced and imported woods, premium ceramics and careful lighting design, tuned to capture soft, indirect daylight while reducing heat gain and glare, give every room a calm, inviting atmosphere from morning to evening.